



Personalized explanations for clinician-AI interaction in breast imaging diagnosis by adapting communication to expertise levels

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ABSTRACT

This paper investigates the impact of personalized AI communication on clinical outcomes in breast cancer diagnosis. Our study examines how different AI communication styles influence diagnostic performance, workload, and trust among clinicians, focusing on imaging diagnosis. We engaged 52 clinicians, categorized as interns, juniors, middles, and seniors, who diagnosed patient cases using conventional and assertiveness-based AI communication. Results show that personalized AI communication reduced diagnostic time by a factor of 1.38 for interns and juniors, and by a factor of 1.37 for middle and senior clinicians, without compromising accuracy. Interns and juniors reduced their diagnostic errors by 39.2% with a more authoritative agent, while middle and senior clinicians saw a 5.5% reduction with a more suggestive agent. Clinicians preferred assertiveness-based AI agents for their clarity and competence, valuing detailed, contextual explanations over numerical outputs. These findings underscore the need for adaptable AI communication to build trust, reduce cognitive load, and streamline clinical workflows. This work offers valuable insights for designing effective AI systems in high-stakes domains, contributing significantly to the Human-Computer Interaction community by enhancing our understanding of AI-mediated clinical support.

1. Introduction

Artificial Intelligence (AI) systems, driven by Deep Learning (DL) techniques, hold promise for various healthcare applications (Calisto, 2024; Hannun et al., 2019; Ruamviboonsuk et al., 2019). Integrating technological advancements in the clinical workflow, such as AI systems, can potentially advance personalized and precision medicine (Ho et al., 2020; Wetzstein et al., 2020). However, these systems often fail to capture the variability among clinicians with different experience levels, such as interns, juniors, middles, and seniors (Uddin et al., 2019). Consequently, understanding how AI systems should communicate, considering clinicians' professional experience, becomes a crucial design question (Pacheco and Martinho, 2019). This study focuses on using the BreastScreening-AI framework (Calisto et al., 2022b) in two scenarios: with conventional and assertiveness-based agents (Pacheco and Martinho, 2019; Paradedda et al., 2019). The

anthropomorphic intelligent agents act as a “second reader” opinion (Ueno et al., 2018), intending to improve diagnostic performance, reducing false positives and negatives, while enhancing the efficiency and efficacy of the clinical workflow.

In high-stakes decision-making environments (Morrison et al., 2023), such as healthcare (Zhang et al., 2024), the level of expertise significantly influences how the *end-users* respond to directives (Seidel et al., 2021). Senior and middle clinicians, with their extensive knowledge and experience, favor communication that respects their autonomy and acknowledges their proficiency, being less receptive to direct orders that might undermine their expertise (Kirmayer et al., 2015). Conversely, intern and junior clinicians benefit from structured and assertive guidance that supports their learning and helps manage cognitive load (Abdelaziz et al., 2020). Given their limited experience, clear and direct instructions are essential for navigating complex tasks.

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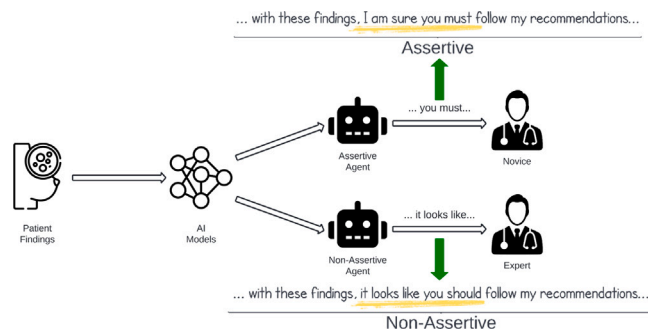


Fig. 1. Communication dynamics between AI agents and clinicians based on the level of assertiveness. The assertive agent communicates with confidence and directness. In contrast, the non-assertive agent uses tentative and less direct language. The communication style adapts to the clinician's experience level to improve diagnostic performance, trust in the AI system, and overall workflow efficiency.

In psychology and cognitive science, assertive communication is defined by clarity, confidence, and directness, while non-assertive communication is more tentative and indirect (Paradedda et al., 2019). When aligning the AI agent's communication style with the clinician's experience level (Fig. 1), it can significantly enhance medical assessments by improving workflow efficiency, ensuring diagnostic accuracy, and building trust (Hohenstein and Jung, 2020). This alignment can also reduce cognitive workload, allowing clinicians to make more informed and confident decisions.

Effective communication improves the reliability and usability of intelligent agents in clinical settings (Hohenstein and Jung, 2020). It is crucial in environments where diagnostic accuracy and clinician trust are critical (Liu et al., 2023). The tone of AI communication significantly shapes user perceptions and responses. It also influences adopting AI systems in medical environments (Pacheco and Martinho, 2019).

Marked by directness and clarity, assertive communication supports interns and junior clinicians by instilling confidence and providing structured guidance in complex decisions (Abdelaziz et al., 2020). Interns benefit from directive support in navigating diagnostic complexity, while juniors gain from clear instructions that reinforce their expertise.

In contrast, more experienced clinicians at middle and senior levels respond best to non-assertive, supportive communication that respects their autonomy and expertise. Middle-level clinicians appreciate communication acknowledging their proficiency, while senior clinicians prefer succinct, relevant information supporting efficient decision-making (Zhang et al., 2024). These differences underscore the need for adaptive AI communication styles aligned with clinicians' expertise levels to maximize AI efficacy and user acceptance in clinical workflows (Ahn et al., 2023; Ho et al., 2020).

A tailored communication approach creates an accessible, intuitive AI experience across expertise levels, enhancing User Experience (UX) by fostering a supportive, user-centric environment (Ahn et al., 2023). This approach facilitates smoother AI integration into clinical workflows and builds satisfaction and trust among healthcare professionals (Calisto, 2024; Paradedda et al., 2019). Effective communication promotes AI adoption in clinical practice by improving interactions and long-term trust, empowering clinicians to make accurate, informed decisions.

Developing intelligent systems that are both technically robust and user-centered is essential for complex medical contexts (Nazar et al., 2021). Our work emphasizes that personalization in AI involves tailoring predictions to patient data and adapting communication to the clinician's expertise level. This approach aligns with clinical needs, as radiologists and healthcare professionals require AI systems that offer

targeted, context-appropriate guidance, enhancing decision accuracy and confidence in the technology.

This led us to the main research question driving this study:

How do different levels of AI assertiveness affect clinicians' medical assessments and perceptions, including efficiency, diagnostic accuracy, trustworthiness, and workload?

Our CHI 2023 study examined the benefits of customized communication between novice and expert clinicians (Calisto et al., 2023a). It explored how personalized AI interactions build trust and engagement. The current work extends this by focusing on the needs of clinicians at different expertise levels. Rather than a novice vs. expert comparison, it analyzes how assertiveness-based AI communication affects interns, juniors, middles, and seniors, providing detailed insights into tailoring AI communication for clinical workflows.

Assertiveness-based communication was introduced as a conceptual model to improve Clinician-AI Interactions. This study advances it by examining how communication style adjustments affect diagnostic performance across expertise levels. Evaluating specific groups reveals their distinct needs. The findings offer insights for designing AI systems that adapt to clinical environments, highlighting tailored communication to improve clinician performance and align AI with expertise.

Our within-subject study involved 52 clinicians diagnosing a dataset of 289 patients. Each clinician interacted with three intelligent agents: (1) conventional; (2) non-assertive; and (3) assertive. The non-assertive and assertive agents were grouped under the assertiveness-based condition to compare the effects of tone variability within the same framework. This approach evaluated how communication strategies affected diagnostic accuracy, efficiency, and trust in AI, offering insights into tailored communication in clinical decision-making.

Unlike the previous CHI 2023 study (Calisto et al., 2023a), which emphasized high-level principles, our current work focuses on measurable outcomes in more detailed expertise groups, such as interns, juniors, middles, and seniors. The results include a more detailed analysis of the diagnostic accuracy and user satisfaction across these detailed levels of clinical expertise. The assertiveness-based agent used varying tones and provided human-interpretable clinical explanations. This approach allowed us to capture the nuanced effects of assertiveness-based communication on clinician perception and effectiveness (Aldoj et al., 2020).

Our findings refine communication strategies for personalized AI in clinical workflows, addressing expertise-level nuances. The study provides practical recommendations for integrating adaptable AI into clinical settings, considering diverse clinician decision-making. These contributions support design guidelines to improve usability, build trust, and enhance AI effectiveness in high-stakes domains.

To summarize, our work advances Human-Computer Interaction (HCI) by showing how adaptive AI communication supports clinician needs in real-world clinical environments. It bridges the gap between conceptual models of assertiveness-based communication introduced at the CHI 2023 conference (Calisto et al., 2023a) and practical healthcare applications (Morrison et al., 2023), offering insights essential for integrating AI into clinical practice.

2. Research questions & hypotheses

Given the impact of communication styles on clinician interactions with AI systems, it is crucial to investigate how assertiveness-based communication influences medical assessments. This work explores the relationship between the AI agent's communication style and the clinician's experience level, focusing on how tailored communication can enhance diagnostic accuracy, workflow efficiency, and trust. Understanding these dynamics will provide insights into designing more effective and user-centered AI systems in healthcare. Such insights are instrumental in bridging the gap between technical capabilities

and user-centered design, ensuring AI integration enhances clinical outcomes and user satisfaction.

The final purpose of our work is twofold. Through assertiveness-based agents, we first aim to understand how personalized communication could affect medical assessments regarding the efficiency and efficacy of the clinical workflow. Secondly, we aim to understand how clinicians perceive assertiveness-based agents differently. Thus, our work addresses two primary research questions concerning the impact of assertiveness-based agents on efficiency and efficacy (RQ1) and clinicians' perception (RQ2). Addressing these questions is critical for advancing AI systems that improve diagnostic outcomes and integrate effectively into diverse healthcare environments, supporting clinicians in making more informed and confident decisions.

Specifically, we consider the following research questions and related hypotheses:

- **RQ1.** What is the impact of an assertiveness-based agent on medical diagnosis?
 - **H1.1.** Clinician efficiency, measured by the time taken per diagnosis, will improve with an assertiveness-based agent.
 - **H1.2.** Clinicians' classification accuracy will remain consistent or improve when using an assertiveness-based agent compared to a conventional agent.
 - **H1.3.** The accuracy variation between novice and experienced clinicians will be influenced by the communication tone level of the agent.
- **RQ2.** What are clinicians' perceptions of an assertiveness-based agent?
 - **H2.1.** Clinicians will show a preference for an assertiveness-based agent.
 - **H2.2.** An assertiveness-based agent will be perceived as more trustworthy by clinicians.
 - **H2.3.** Tailored explanations and highlights will not add to clinicians' workload nor reduce usability.
 - **H2.4.** Perceptions of reliability and capability will differ between novice and experienced clinicians based on their communication tone levels.

Recent studies emphasize the importance of communication tone in human-AI collaboration. Morrison et al. (2024) explored the effects of imperfect eXplainable-AI (XAI) on decision-making, highlighting the need for clarity and contextualized explanations. Similarly, Kim et al. (2024) examined how uncertainty expression in large language models impacts user reliance and trust. Our work builds on these findings by focusing on assertiveness-based communication tailored to clinicians' expertise, comparing its efficacy and perceptions with conventional approaches.

In real-world clinical settings (Su et al., 2022), clinicians' time is a limited and expensive resource that should be reallocated efficiently. We take the position that while clinicians should make their clinical assessments with care, AI agents can help improve diagnosis efficiency. Our assertiveness-based agent is designed to summarize the recommendations and provide clinical arguments explaining the underlying classification and segmentation of the AI models, bridging the gap between AI interpretability and clinician needs for actionable insights.

3. Background

In this section, we explore the role of breast imaging in detecting and diagnosing breast cancer, emphasizing the need for precision and timeliness in patient care. Variability in imaging quality and interpreting subtle features, particularly in dense or ambiguous cases, pose challenges that require clinical expertise. AI-driven tools address these issues by extending radiologists' capabilities with tailored, context-sensitive recommendations. These advancements support the shift towards data-driven, personalized healthcare, enabling radiologists to navigate complex scenarios and improve diagnostic workflows.

3.1. Clinical relevance of imaging modalities

The AI models in this study were trained on data from the following list of three key imaging modalities. Each modality was independently analyzed to leverage its unique diagnostic strengths (Appendix A.3), ensuring a comprehensive assessment of breast tissue (Calisto, 2017). This approach allowed the models to integrate insights across modalities, enhancing diagnostic accuracy and addressing the limitations of single-modality systems.

Each modality provides unique insights:

1. **MammoGraphy (MG):** Widely used for routine screening, MG excels in detecting calcifications and structural distortions. However, its sensitivity is limited in dense breast tissue.
2. **UltraSound (US):** Effective in distinguishing solid masses from cysts, US is indispensable for guiding biopsies and examining dense breast tissue.
3. **Magnetic Resonance Imaging (MRI):** With superior soft-tissue contrast, MRI is highly sensitive for detecting invasive cancers, particularly in high-risk patients.

The AI system capitalizes on its strengths while addressing its limitations by independently analyzing each modality. This modality-specific approach reduces diagnostic errors and enhances the precision of severity predictions, ensuring that AI recommendations align closely with clinical practice.

3.2. Diagnostic imaging in breast cancer

Breast cancer diagnosis relies on advanced imaging modalities such as MG, US, and MRI. Each modality offers unique advantages, contributing distinct diagnostic insights essential for comprehensive patient care. However, imaging quality variability and the nuanced interpretation of subtle or ambiguous features remain significant challenges. This complexity often demands expert clinical judgment, particularly in cases involving dense breast tissue or borderline imaging findings.

AI-driven diagnostic tools have emerged as valuable allies in addressing these challenges. In fact, by complementing the expertise of radiologists (Yildirim et al., 2024a), these systems provide context-sensitive recommendations that integrate seamlessly with clinical workflows. The incorporation of AI into diagnostic processes signifies a transformative shift towards data-driven and personalized healthcare. This study builds upon this foundation to explore how AI communication strategies can enhance clinical decision-making in breast cancer diagnosis.

3.3. Rationale for severity prediction

The Breast Imaging Reporting and Data System (BIRADS) is a standardized framework used globally to report breast imaging findings and guide clinical management (Spak et al., 2017). Unlike predicting cancer directly, this study focuses on BIRADS prediction to align AI outputs with the decision-making processes of clinicians (Appendix A.1). BIRADS offers structured categories that encapsulate both diagnostic findings and subsequent management recommendations, bridging the gap between imaging results and clinical actions.

Clinicians use BIRADS scores to evaluate the likelihood of malignancy and determine appropriate next steps, such as follow-ups or biopsies. Predicting BIRADS values allows AI systems to provide actionable insights that radiologists can directly incorporate into their workflows (Appendix A.7). Moreover, the BIRADS framework reflects clinical uncertainty through its probabilistic categories, offering a nuanced lens for studying how AI communication impacts trust and decision-making confidence across varying expertise levels.

3.4. Addressing ground truth ambiguities

The ground truth for this study is established using BIRADS scores provided by the head of radiology, an expert clinician with extensive experience at the Hospital Professor Doutor Fernando Fonseca (HFF), EPE clinical institution. Such scores are informed by a comprehensive understanding of patient cases (Appendix A.2), incorporating imaging data and supplementary information like biopsy confirmations or long-term follow-up outcomes. This dual-source approach enhances the reliability of the ground truth, aligning annotations more closely with real-world diagnostic accuracy.

The framework does not assess how borderline cases impact ground truth reliability (Appendices A.5 and A.6). Cases between BIRADS 3 and 4 pose significant challenges in clinical practice. Future work should examine AI performance in such uncertain scenarios by using probabilistic or confidence metrics in annotations. This approach could enhance understanding of clinician-AI interaction dynamics and the potential for personalized AI support in ambiguous cases. Incorporating multi-clinician input and probabilistic scoring could enhance the robustness and scalability of AI diagnostics, aligning them with clinical complexities.

3.5. Tackling ambiguities in BIRADS categories

The inherent ambiguity in BIRADS categories, particularly in borderline cases, underscores the need for tailored AI support. Such cases, where distinctions between categories like BIRADS 3 and 4 are subtle, often challenge even experienced radiologists, potentially leading to variability in interpretation and patient management. Incorporating AI systems capable of providing probabilistic metrics and confidence scores can help clinicians make more informed decisions, bridging the gap between diagnostic uncertainty and actionable insights.

Key challenges include:

- Probable Benign (BIRADS 3): Findings often warrant short-interval follow-ups, which can introduce uncertainty for both clinicians and patients.
- Suspicious Abnormality (BIRADS 4): Subdivided into low, moderate, and high suspicion (4A, 4B, 4C), this category spans a wide probability range, altering biopsy decisions.

The AI system mitigates these challenges by offering personalized recommendations based on clinician expertise levels. For example, junior clinicians benefit from directive AI guidance, while senior clinicians receive collaborative suggestions. This adaptability fosters trust and enhances decision-making confidence, particularly in ambiguous cases where clinical judgment is paramount.

3.6. Integrating AI models with clinical practice

This section explores how advanced AI models enhance clinical workflows by tailoring outputs to each imaging modality. The modular design provides precise, interpretable insights that complement clinicians' expertise, promoting transparency and trust essential for adoption in healthcare settings.

The proposed system employs state-of-the-art AI models tailored to the specific requirements of each imaging modality:

- DenseNet for 2D imaging (MG and US) to ensure efficient feature propagation and mitigate vanishing gradients (Huang et al., 2019).
- 3D ResNet for volumetric MRI data to handle complex spatial information and provide robust predictions (Aldoj et al., 2020).

Each model generates a modality-specific BIRADS score, preserving the integrity of the diagnostic process. This design emphasizes modularity and flexibility (Appendices A.4 and A.8), allowing clinicians to review AI-generated insights for individual modalities before integrating them into their final assessment. Such a framework aligns with HCI principles, promoting transparency, trust, and usability in high-stakes environments.

4. Related work

In breast cancer, medical imaging systems enable *end-users* to diagnose various modalities, including MG, US, and MRI, through seamless data retrieval (Calisto et al., 2023a). Integrating these modalities presents opportunities for quantitative imaging and diagnoses, necessitating specialized data handling, post-processing, and visualization methods (Igarashi et al., 2016). Multimodal imaging not only enhances diagnostic accuracy by leveraging the strengths of each modality (Yildirim et al., 2024a), but also supports personalized medicine, tailoring treatments based on individual patient characteristics.

Integrating diverse medical imaging modalities into AI systems demands advanced algorithms (Bi et al., 2019; Panayides et al., 2020). These algorithms must process varied data types, enable comprehensive analyses, and support improved clinical decision-making. Tailored communication strategies make AI outputs accessible and intuitive for clinicians. Such strategies enhance the UX and build trust in AI-assisted diagnostics, ensuring the systems effectively support clinical workflows.

Medical imaging tools aid experts in identifying cancer prognostics from multimodal data (Yildirim et al., 2024a). Leveraging the strengths of diverse imaging modalities enhances decision-making and provides deeper insights into complex diagnostic challenges (Calisto et al., 2020). To that end, we focus on understanding different aspects and expectations of a Clinical Decision Support System (CDSSe) integrated into the radiology workflow, highlighting the enhancement of the multimodality approach to medical imaging diagnosis through assertiveness-based interaction.

Recent studies on anthropomorphic agents reveal their dual impact on user interaction (Bi and Huang, 2023; Inie et al., 2024). Human-like traits in AI can enhance trust and engagement by fostering familiarity and aligning communication with user preferences and expertise (Chandio et al., 2024). However, challenges like over-reliance and misinterpreting AI capabilities arise in high-stakes environments (Bućinca et al., 2021; Gade et al., 2020). Balancing these trade-offs requires selective integration, ensuring transparency while enhancing usability. Our approach incorporates anthropomorphic elements to support assertiveness-based interaction, aligning communication with clinician needs without compromising trust or interpretability.

4.1. Communication theory

Assertiveness-based communication involves confidently and directly conveying information while maintaining respect for the listener's autonomy and preferences (Morrison et al., 2024; Paradedda et al., 2019). From behavioral psychology (Ames et al., 2017), assertiveness-based has been extensively studied in contexts such as conflict resolution and team collaboration (Ashraf and Asad, 2021; Babel et al., 2022), where effective communication is central to achieving desired outcomes.

In the HCI context (Quinn and Zhai, 2016; Wong et al., 2019), assertiveness-based plays a dual role. On the one hand, assertive-based communication enhances trust by projecting confidence and authority (Chang et al., 2024), particularly in systems designed to assist with high-stakes decision-making like clinical diagnostics. On the other hand, overly assertive-based communication risks alienating users (Paradedda et al., 2020), especially those with less experience or lower confidence in their decision-making abilities. These trade-offs

underline the need for personalized communication strategies tailored to the user's expertise level.

Recent work highlights the importance of balancing assertiveness-based with adaptability in XAI (Morrison et al., 2024). For example, assertiveness-based AI recommendations have been shown to influence user reliance and decision-making effectiveness. Systems that modulate assertiveness-based behaviors on user expertise can mitigate cognitive overload in novice users while maintaining efficiency for expert users (Calisto et al., 2023a).

Our study builds on this foundation by exploring assertiveness-based communication within the context of personalized CDSSes. We examine how tailored communication tones, ranging from assertive to non-assertive, impact diagnostic accuracy and trust across clinicians with varying levels of expertise. Unlike prior work that primarily investigates assertiveness-based communication in general settings (Morrison et al., 2024; Paradedda et al., 2019), our approach integrates it as a core design principle for enhancing clinician-AI collaboration in breast imaging diagnostics. The results highlight the potential of assertiveness-based strategies to address cognitive biases, streamline decision-making, and improve the interpretability of AI-generated insights.

4.2. Human-AI interaction

In the Human-AI Interaction (HAI) context, intelligent agents must go beyond providing results alone and consider user behaviors during decision-making (Xie et al., 2020). We explore deeper into this HAI literature, exploring the interdisciplinary topic of XAI, which intersects cognitive psychology (Yao, 2021), learnability (Abdul et al., 2018), and context awareness (Dourish et al., 2020) within the field of HCI (Shneiderman, 2016).

Effective XAI techniques ensure AI systems are understandable (Schneeberger et al., 2023), interpretable (Saranti et al., 2022), and user-friendly (Evans et al., 2022), particularly in complex domains like healthcare (González-Alday et al., 2023; O'Sullivan et al., 2022; Santiago et al., 2022). Addressing these aspects bridges the gap between AI technology and practical, user-centric applications (Lombardi et al., 2024), enhancing user trust and decision-making efficiency in clinical environments (Saßmannshausen et al., 2024).

Recent studies highlight the importance of personalized communication in XAI. For instance, Cognitive Forcing Functions (CFFs) have been shown to reduce overreliance on AI by compelling users to engage critically with explanations (Buçinca et al., 2021). However, interventions like CFFs often involve trade-offs, such as increased complexity and reduced usability. Other works emphasize that tailoring XAI explanations to individual user characteristics, such as age or cognitive style, can influence engagement and trust (Nimmo et al., 2024). These findings suggest that overly granular personalization may not yield better outcomes, and that focusing on broader user groups may be more effective.

Cognitive psychology plays a vital role in understanding how users interact with AI systems (Yao, 2021), particularly in perceiving, interpreting, and responding to AI-generated information. Learnability, an essential usability aspect, encompasses various elements such as hints, guidance, and visualizations when designing XAI systems (Abdul et al., 2018; Amann et al., 2020). Furthermore, explainable context awareness simplifies context representation, providing users with information about the obtained data and system actions (Dourish et al., 2020). These aspects collectively contribute to making AI systems more user-friendly and interpretable.

HAI incorporates human feedback in model training to create better ML models (Aha and Coman, 2017; Amershi et al., 2019; Calisto et al., 2017; Kocielnik et al., 2019), where we bring the topic focusing on personalized and customized AI suggestions tailored to varying levels of medical expertise. Researchers emphasize the importance of explaining AI systems' reasoning to enhance HAI, particularly in

medical domains, where the interpretability of AI predictions is crucial (Bansal et al., 2021; Kawamleh, 2022; Rudin, 2022). However, these approaches often fail to account for cognitive bias in decision-making, which varies among individuals with different levels of expertise and knowledge (Abdelaziz et al., 2020; Seidel et al., 2021). Our work addresses these gaps by studying assertiveness-based communication in AI systems, considering expertise levels to reduce cognitive bias.

4.3. Clinical decision-making

Clinical decision-making involves complex cognitive processes in which clinicians synthesize vast amounts of data to make informed decisions about patient care (Bansal et al., 2021; Lombardi et al., 2024; Saßmannshausen et al., 2024). This process requires not only the integration of clinical knowledge and patient information (Alowais et al., 2023; Jacobs et al., 2021), but also the application of judgment under conditions of uncertainty (Morrison et al., 2024; Schaeckermann et al., 2020; Sivaraman et al., 2023; Yildirim et al., 2024b). Effective decision-making in clinical settings depends on the ability of the clinician to interpret diagnostic information accurately (Lee et al., 2021), recognize patterns (Nazar et al., 2021), and weigh potential outcomes against each other (Calisto, 2024).

Integrating AI into clinical decision-making involves challenges and unintended consequences (Miller, 2019; Yang et al., 2023a,b). Patient safety, clinician fatigue, and medical errors require careful attention (Alexander et al., 2022; Sharma et al., 2022; Sittig et al., 2022; Wang et al., 2021). Many clinicians struggle with AI systems due to limited technical skills and insufficient customization to their needs (Calisto et al., 2022a).

Poorly designed interfaces limit clinicians' ability to understand AI outcomes effectively (Edgar et al., 2022). This challenge also impacts their ability to communicate these outcomes. Many interfaces do not account for differences in clinician characteristics during decision-making. For example, novice and expert clinicians often use distinct reasoning approaches, which are frequently overlooked in interface design.

Current studies often overlook how expertise shapes clinicians' interpretation of AI-generated information (Seidel et al., 2021; Yao, 2021). This gap is critical in radiology, where diagnostic decisions carry serious consequences (Hanna et al., 2018; Seely and Alhassan, 2018). Overdiagnosis, or false positives, leads to unnecessary biopsies and treatments, causing patient stress and straining resources. Underdiagnosis, or false negatives, delays critical interventions, jeopardizing outcomes. Tailored AI communication strategies are vital to reducing cognitive biases and improving accuracy.

Our work examines how diagnostic accuracy varies with expertise, focusing on false positives and negatives among interns, juniors, middle-level, and seniors. With less experience, interns and juniors are more prone to false negatives, often missing critical findings. Middle-level clinicians balance AI recommendations with their judgment, reducing errors. Senior clinicians benefit from suggestive AI communication, which helps minimize false positives. Tailoring AI communication to expertise levels mitigates cognitive biases and enhances diagnostic accuracy, providing insights for improving AI integration in healthcare.

4.4. Healthcare challenges in deploying AI systems

The lack of large-scale deployment of AI systems in healthcare further complicates understanding how these systems are perceived and used in real-world settings (Su et al., 2023; Wang et al., 2021; Zappatore et al., 2023). Interactive Machine Learning (iML) (Lee et al., 2021), Human-In-The-Loop (HITL) (van der Stappen and Funk, 2021), human-AI symbiosis (Jarrahi, 2018), and Human-AI Collaboration (Wang et al., 2021) are approaches proposed within HCI to address

these challenges. Integrating AI systems into healthcare requires advanced algorithms to process diverse data types and provide personalized recommendations. Understanding clinical personalization and its impact on AI in healthcare can bridge knowledge gaps, foster trust, and enhance AI adoption in clinical practice.

Improving prediction accuracy, model efficiency, and interpretability often neglects the operational burden on healthcare professionals (van der Stappen and Funk, 2021; Su et al., 2022). Studies on AI systems' perception and usage in clinical decision-making frequently overlook differences in behavioral reasoning (Larasati, 2022). Furthermore, some methods prioritize accurate algorithmic suggestions without considering the clinician's professional experience (Panigutti et al., 2022).

CDSSes have greatly benefited from DL algorithms (Esteve et al., 2019) in various applications. DL systems can detect patterns, make predictions, and assist clinicians in high-stakes decision-making processes (Su et al., 2022), such as skin cancer diagnosis (Esteve et al., 2017), cardiac MRI segmentation (Medley et al., 2019), and breast cancer detection (Maicas et al., 2019). However, these algorithms' "black-box" nature poses challenges regarding transparency and trust, which are critical for their adoption in clinical settings.

Enhancing explainability and user-centric design shows promise for improving AI integration (Donoso-Guzmán, 2023), particularly in high-stakes domains like healthcare. While AI models excel at identifying patterns in medical data and sometimes surpass human performance (McKinney et al., 2020; Raghu et al., 2019), the impact of explainability remains uncertain Babic et al. (2021). Evidence has yet to conclusively link explainability to greater acceptance, trust, or effectiveness of AI systems (Markus et al., 2021; Shin, 2021). Future research should target scenarios where explainability provides measurable value for adoption and user satisfaction.

Researchers suggest that meaningful explanations could address trust barriers and support reasoning in critical decision-making. However, many AI systems, such as search engines and predictive maintenance tools, function effectively without detailed explanations, showing that explainability is not universally essential. Identifying where explainability enhances decision-making and user engagement remains a key focus for future research.

The healthcare context introduces unique challenges where clinicians must balance trust in AI recommendations with their expertise (Panigutti et al., 2023). Tailoring AI communication to align with individual behavioral characteristics and experience levels remains crucial (Ahn et al., 2023; Calisto et al., 2022b; Maicas et al., 2019). Personalized strategies may help bridge gaps in understanding and foster better collaboration between clinicians and AI systems, even without robust evidence on explainability's direct impact. Future research should explore the conditions under which explainability enhances decision-making, trust, and system adoption, particularly in scenarios where acceptance may be contingent on understanding AI outputs.

While DL-based CDSSes show promise (McKinney et al., 2020; Rajpurkar et al., 2022), translating them to clinical settings is challenging, often neglecting clinicians' varied cognitive processes and decision-making styles, leading to usability issues. Our research addresses these challenges by tailoring algorithmic suggestions to different levels of medical expertise. Unlike previous work (Calisto et al., 2023a), this study customizes AI communication to align with the distinct cognitive styles of interns, juniors, middles, and seniors. Integrating HCI and AI empowers clinicians, enhancing decision-making, improving patient care, and addressing each clinician category's needs for seamless AI integration.

4.5. Clinical adoption and trust in AI systems

Utility to clinicians and logistical hurdles are common obstacles in clinical adoption (Calisto et al., 2022a), and some systems have not effectively reduced clinician workload (Cai et al., 2019a) or improved

diagnostic accuracy (Kohli and Jha, 2018). HCI research in clinical environments has explored the evaluation of interactive DL systems from a human-centered perspective (Cai et al., 2019b; Park et al., 2019; Shneiderman, 2022). Yet, there is a gap in understanding how to integrate these systems seamlessly into existing workflows without adding to the operational burden of healthcare professionals.

Studies have investigated techniques to enhance diagnostic utility and user trust in DL predictions, as well as identified important information for clinicians when integrating AI assistants into routine practice (Cai et al., 2019a,b). However, these studies have not considered the heterogeneous behavioral nature of decision-making among clinicians. Additionally, the existing literature often overlooks the need for adaptive systems that can cater to varying levels of expertise and diverse cognitive processes among healthcare professionals.

Trust is crucial in clinical environments where life-altering decisions are made (Calisto et al., 2022a). Positive motivational attribution and reducing ambiguity through trust are key factors in successful human-AI collaboration (Bruzzeze et al., 2020; Hohenstein and Jung, 2020; Li et al., 2020; Zagalsky et al., 2021). The impact of assertiveness-based AI mediation on novice and expert clinicians is still poorly understood (Lundberg et al., 2020). Comprehensive studies on how clinician expertise affects the trust and effectiveness of AI-mediated communication are lacking. This gap highlights the need for adaptive AI systems that adjust communication strategies based on the user's expertise and cognitive style.

Understanding the effects of AI communication is crucial to avoid clinical consequences (Sutton et al., 2020). Trust influences clinicians' perceptions of AI outcomes, attitudes, satisfaction, and performance evaluations (Panigutti et al., 2022). Attribution theory and external cues shape clinicians' interpretation of information (Hohenstein and Jung, 2020).

Our work builds upon existing literature by incorporating assertiveness-based communication theories into CDSSes used in clinical settings. A key innovation lies in tailoring AI communication strategies to align with the expertise levels of individual clinicians. Unlike prior studies (Calisto et al., 2023a), our approach focuses on enhancing diagnostic performance through personalized communication that meets the unique needs of clinicians at varying stages of their careers.

5. System

The section presents the assertiveness-based AI communication system evaluated in our user studies. The system incorporates granular patient information into its outputs to ensure contextually relevant and precise recommendations (Fig. 2). It also adapts its communication tone to align with the clinician's level of expertise. Tailored interactions are provided for interns, juniors, middle-level clinicians, and senior practitioners, enhancing usability and engagement across varying experience levels.

AI outputs were designed to deliver detailed clinical arguments and patient-specific lesion details. These outputs were compared against a conventional agent that provided only numeric estimates, such as BIRADS scores and accuracy metrics (Sections 3.3 and 3.4). Such a comparison aimed to evaluate the impact of tailored communication on enhancing diagnostic accuracy and clinician trust. Additionally, the study explored how personalized communication strategies influence clinical workflow efficiency and decision-making confidence.

5.1. Integration of contextual and morphological data

The system's capacity to present contextual and morphological patient data bridges the gap between AI and clinical reasoning. Incorporating detailed patient information (Rao et al., 2016), such as lesion lexical characteristics (e.g., size, location, shape, margin, and density) and patient clinical history (e.g., family and personal histories or other granular information), allows our system to offer a comprehensive perspective for well-informed clinical decisions.

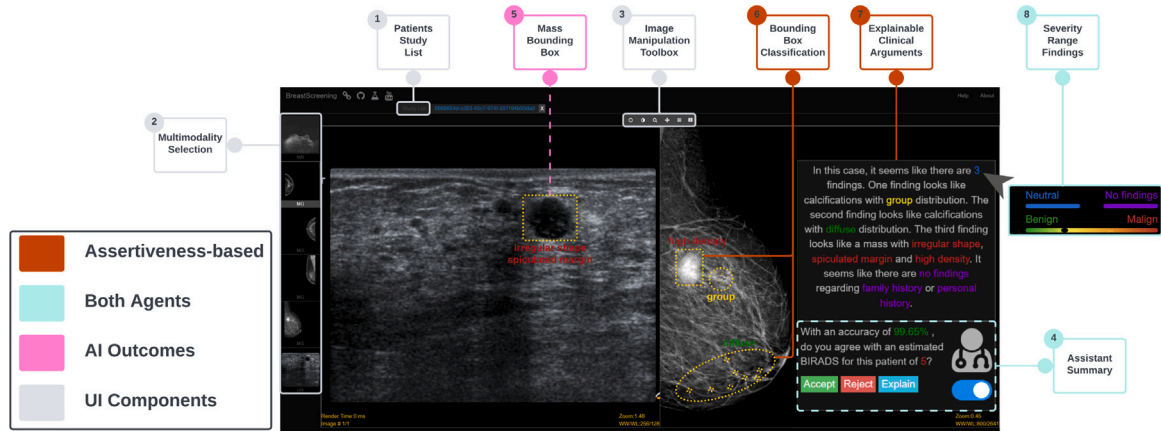


Fig. 2. Interface for assertiveness-based AI agent in medical imaging analysis. The intelligent agent includes several variables to aid clinicians during decision-making. These variables are mimicking clinicians reasoning. Colors indicate severity (green to red) and the number of findings (blue), with personal history variables in purple.

Table 1

The available key functionalities of the radiology UI designed to support diagnostic processes. Functionalities include (△) Analytical Tools, (◊) UI Organizational Mechanism, and (⊕) AI Inferences.

Number	Functionality	Number	Functionality
1 ◊	Patients Study List	10 △	Patient Severity (BIRADS)
2 ◊	Multimodality Selection	11 △	DL Model Accuracy
3 ◊	Image Manipulation Toolbox	12 ◊	Hidden Assistant Toggle
4 ◊	Assistant Summary	13 △	Number of Findings
5 ⊕	Lesion Mass Bounding Box	14 ⊕	Lesion Mass Shape
6 ⊕	Bounding Box Classification	15 ⊕	Lesion Mass Margin
7 ⊕	Explainable Clinical Arguments	16 △	Lesion Mass Density
8 △	Severity Range Findings	17 △	Calcification Patterns
9 ◊	User Choices	18 ◊	Family & Personal Histories

In our particular case of breast diagnosis, contextual data included the description of the two main lesions: (a) masses; and (b) calcifications. Contextual data, including calcification patterns, lesion mass shape, margins, and density, is crucial for accurate diagnosis and treatment planning. Additionally, morphological data visually represents these attributes, enabling clinicians to understand and interpret the information better.

5.2. User interface design

The User Interface (UI) of our anthropomorphic intelligent agent is designed to enhance clinical workflows and support decision-making processes (Fig. 2). The design builds on familiar patterns in clinical practice, aligning with radiologists' existing workflows and expectations. Tools within the interface streamline data retrieval (Table 1), facilitate diagnostic accuracy, and encourage informed decision-making.

During initial assessments, the UI provides quick access to the patient's study list and imaging modality selection (Section 3.1), allowing radiologists to gather necessary data efficiently. Image manipulation tools enable detailed examination and adjustments, helping radiologists explore specific areas of interest. As users progress through tasks, the assistant summary and AI-generated clinical arguments complement the radiologist's analysis, offering additional perspectives while maintaining their decision-making authority.

The functionalities presented in Table 1 are grouped into three distinct categories: (1) Analytical Tools; (2) UI Organizational Mechanisms; and (3) AI Inferences. Analytical Tools (△) include Patient Severity (BIRADS) and Deep Learning Model Accuracy, which provide metrics to support decision-making. UI Organizational Mechanisms (◊) streamline navigation and data management through features like the Patient Study List and Multimodality Selection. AI Inferences (⊕) are designed to use ML techniques to generate insights, including the

Lesion Mass Bounding Box and Explainable Clinical Arguments. These categories, developed through our co-design approach (Section 6.1), form a user-centered framework that optimizes diagnostic workflows.

The UI incorporates feedback from clinicians gathered during co-design workshops. Our design prioritizes alignment with their practical needs and preferences, ensuring the UI supports real-world clinical workflows. To avoid relying on assumptions, established conventions from existing clinical tools guide the design process. This approach minimizes cognitive load and enhances usability by offering a seamless and intuitive UX.

The UI is designed to provide an intuitive and efficient experience, enabling clinicians to navigate the system seamlessly and without interruptions. Interactions are intentionally structured to align with familiar workflows, reducing learning curves and minimizing distractions. Upcoming evaluations will focus on determining the UI's effectiveness in enhancing workflow efficiency, facilitating informed decision-making, and ultimately improving the quality of patient care.

The (1) *Patients Study List* allows clinicians to manage and access patient data efficiently, streamlining the review process with quick navigation and retrieval of records. The (2) *Multimodality Selection* enables switching between imaging modalities like MG, US, and MRI, providing a comprehensive analysis from multiple perspectives and enhancing diagnostic capabilities by correlating findings. The (3) *Image Manipulation Toolbox* offers tools for adjusting and annotating images, including zooming, rotating, highlighting areas of interest, and adding annotations and measurements.

The (4) *Assistant Summary* concisely overviews patient status and suggested actions. The (5) *Lesion Mass Bounding Box* and (6) *Bounding Box Classification* highlight regions of interest for accurate lesion identification. The (7) *Explainable Clinical Arguments* clarifies the AI's decision-making process, aiming to strengthen clinician confidence and understanding. The (8) *Severity Range Findings* visually present abnormality severity levels. These tools enhance clarity, guide attention to critical areas, support accurate assessments, and facilitate effective communication.

The (9) *User Choices* functionality allows clinicians to accept or reject AI suggestions (Fig. 3), ensuring control over final decisions. Further enhancing the system, the (10) *Patient Severity (BIRADS)* displays standardized assessments of lesion severity, while (11) *DL Model Accuracy* communicates the performance metrics of the AI models, reinforcing transparency. The (12) *Hidden Assistant Toggle* enables users to display or hide the AI assistant as needed. Additionally, the interface includes functionalities, such as the (13) *Number of Findings*, (14) *Lesion Mass Shape*, (15) *Lesion Mass Margin*, (16) *Lesion Mass Density*, and (17) *Calcification Patterns*, providing detailed morphological and lexical data critical for diagnosis (Rao et al., 2016). Finally, including (18) *Family & Personal Histories* ensures a holistic view of the patient's background, aligning with personalized medicine approaches.

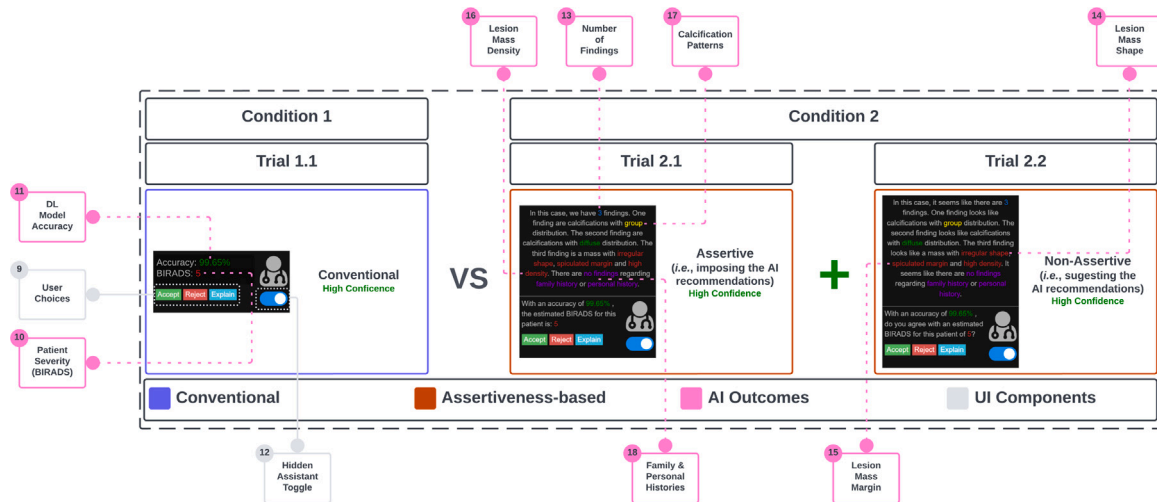


Fig. 3. The two conditions of this study, where clinicians interact: Condition 1 – Conventional Intelligent Agent; and Condition 2 – Assertiveness-based Intelligent Agent. Each clinician interacted with three trials: Trial 1.1 – Conventional; Trial 2.1 – Assertive Communication, imposing the AI recommendations to clinicians; and Trial 2.2 – Non-Assertive, suggesting the AI recommendations to clinicians.

5.3. Co-design with clinicians

The UI of our intelligent agents was co-designed with the clinicians (Section 6), ensuring that the order and presentation of the attributes reflect their needs and preferences (Fig. 2 and Fig. 3). Initially, our BreastScreening-AI framework was developed without controls for user feedback (Calisto et al., 2020), which also limits the DL model in learning from the feedback provided actively by clinicians (Ren et al., 2021). After testing and feedback sessions with clinicians (Calisto et al., 2017, 2021), we incorporated functionalities for acceptance, rejection, and explanations of AI decisions. These enhancements made the AI system more responsive and adaptable to clinicians' needs.

The UI features a layout with transparent sections for functionalities, enhancing accessibility and minimizing cognitive load (Section 7). Controls enable clinicians to interact with AI recommendations (Calisto et al., 2022b), granting autonomy to accept or reject suggestions. This fosters a dynamic, collaborative environment, positioning AI as an assistive tool rather than an authoritative entity. The design promotes user engagement and trust by allowing clinicians to use AI insights while keeping control over final decisions.

AI-generated explanations increase transparency by emulating clinicians' reasoning patterns in their decision-making processes. These explanations provide clear, interpretable details, fostering trust in the system. The UI is designed with a user-centered approach, ensuring seamless integration into clinical workflows. Its intuitive design supports clinicians by enhancing both diagnostic efficiency and user satisfaction. Continuous feedback from users, coupled with iterative refinements, ensures the system evolves to meet the ever-changing demands of clinical practice.

In summary, the system UI was shaped by co-design workshops and clinician feedback. Our design approach and developments prioritized usability, aligning with clinical workflows. This alignment helps to minimize cognitive load, facilitating seamless integration into everyday diagnostic tasks. Visual tools, including severity indicators and explainable clinical arguments, were incorporated to improve clarity and strengthen trust in the AI-generated outputs. These features aim to support clinicians in making informed, confident decisions while enhancing overall workflow efficiency.

5.4. Comparison of AI agents

Our assertiveness-based agent uses recommendations for classifying and segmenting: (i) the number of detected findings; (ii) the patient

severity of each breast and per medical imaging modality; (iii) a visual scale representing the benign or malignant estimates; (iv) providing visualization of the sensitivity and specificity outcomes of the models; and (v) with clinical arguments of the patient, such as pathological co-variables. To compare¹ the assertive and non-assertive agents with the conventional agent (Fig. 3), participants were informed that the AI models generate recommendations to evaluate performance.

The above Fig. 2 illustrates how the two AI agents were integrated into an existing medical workflow to classify medical data supporting breast cancer diagnosis. Both agents are recommending classification based on the DenseNet model (Huang et al., 2019) for MG and US, as well as based on the ResNet model (Morais et al., 2023) for MRI volumes (Section 3.6). The two AI agents provided the severity classification of the patient via BIRADS (Spak et al., 2017) and the model's accuracy for that classification.

Classifications use radiologist-annotated lesion segmentations to improve interoperability. Highlighted regions of interest focus attention on critical areas influencing AI classifications. This visual guidance clarifies the logic of the DenseNet and ResNet models, supporting decision-making. The approach builds trust in the agent's reliability by aligning AI outputs with clinical reasoning.

6. Methodology

We examined how personalized communication impacts AI assistance, focusing on customized tones in clinician-agent interactions. From March to June 2022, 52 semi-structured interviews and user tests were conducted to explore clinician needs and preferences. Input from two ML and three HCI experts refined the design and methodology, ensuring a comprehensive, user-centered approach.

Healthcare professionals participated in the study under approved ethical protocols. Clinicians and researchers engaged in interactive focus groups, offering perspectives that enriched the process. Their input shaped the intelligent agents' design to align with real-world clinical practices.

¹ For more details, the [prototype-assertive-proactive](https://github.com/Calisto-FM/prototype-assertive-proactive) repository gives access to the assertive prototype (t.ly/Fx7Pj), the [prototype-non-assertive-proactive](https://github.com/Calisto-FM/prototype-non-assertive-proactive) repository to the non-assertive prototype (t.ly/RXHnq), and the [prototype-multi-modality-assistant](https://github.com/Calisto-FM/prototype-multi-modality-assistant) repository to the conventional prototype (git.io/JMjDi). Last accessed July 7, 2024.

To evaluate the impact of customized communication on clinical decision-making, we implemented a within-subjects experimental design. Participants completed three trials, interacting with both conventional and assertiveness-based communication styles (Fig. 3). The assertiveness-based condition included both assertive and non-assertive styles, allowing clear comparisons of their effects on clinician performance and decision-making.

The conventional AI condition served as a baseline, offering diagnostic recommendations without explanation. Assertiveness-based conditions included explanations tailored to clinicians' expertise. This design allowed comparisons between baseline performance and the effectiveness of assertive and non-assertive communication styles, addressing both general efficacy and interaction dynamics.

Our co-design approach played a central role in shaping the prototypes. Collaborative sessions provided critical insights into clinician preferences, particularly on assertiveness levels in agent communication. These sessions refined the design to meet clinical needs and user expectations, ensuring a user-centered and effective solution.

6.1. Co-design process and iterative development

The co-design process was central to the system's development, enabling collaboration with clinicians at every stage. Workshops, interviews, and pilot testing provided insights that guided iterative design decisions. This ensured the system's functionalities and communication styles aligned with clinical workflows while accommodating diverse expertise levels.

To ensure effective participation, we assessed the availability of 52 clinicians for semi-structured interviews, pilot testing, and formative user studies. Due to the time-intensive nature of these activities, 16 clinicians participated in semi-structured interviews, pilot testing, and formative user studies. The remaining 36 clinicians focused solely on the semi-structured interviews and formative user studies, as this better suited their schedules and interests. This careful planning ensured balanced and comprehensive engagement across all stages of the co-design process.

6.1.1. Focus groups & workshops

Focus groups and workshops were key in shaping the system's design. These sessions involved groups of 4–6 clinicians from diverse clinical settings, including public hospitals, cancer institutes, private clinics, and the two ML experts and three HCI experts. Discussions explored clinicians' preferences for AI communication, usability, and decision-making workflows. Each session was conducted remotely using secure video conferencing platforms, which lasted approximately 60 min, fostering a collaborative environment for sharing insights and addressing contrasting perspectives.

Key usability challenges within the system were identified through collaborative discussions in focus groups. Participants discussed practical difficulties and shared insights that shaped the refinement of essential features. A particular emphasis was placed on enhancing user control and ensuring transparency in AI interactions. Severity indicators visually representing clinical findings' urgency or seriousness were crucial for effective decision-making. Explainable clinical arguments, designed to clarify the reasoning behind AI-generated outputs, also emerged as indispensable elements during these collaborative sessions.

6.1.2. Semi-structured interviews

Parallel to the workshops, 52 one-on-one semi-structured interviews allowed an in-depth exploration of individual clinicians' experiences and expectations. Each 30–45-minute interview followed a flexible protocol to elicit detailed feedback on interaction styles, preferences for integrating patient-specific data, and reactions to early prototypes. The interviews uncovered nuances in clinicians' workflows and tailored the system's functionalities to their specific needs.

6.1.3. Pilot testing and early feedback

Pilot testing involved 16 clinicians (i.e., 15 radiologists, and 1 breast surgeon), who interacted with early prototypes during April and May 2022. The goal was to evaluate usability and refine critical system components before formal formative user studies. Participants completed diagnostic tasks while providing feedback on functionality, visual clarity, and overall ease of use. Insights led to adjustments such as refining the "Lesion Mass Bounding Box" functionality, enhancing severity range visuals, and improving navigation tools to align better with clinical workflows.

6.1.4. Integration and iterative refinement

The system was refined iteratively using feedback from workshops, interviews, and pilot testing. Clinicians emphasized integrating patient-specific data, enabling detailed contextual information like lesion characteristics, family history, and clinical severity. Customizable communication tones were also prioritized to adapt recommendations to clinicians' expertise, from interns to senior practitioners.

Detailed suggestions from pilot testing led to significant updates, including enhanced visual clarity in the "Lesion Mass Bounding Box" and more intuitive severity range visuals. Navigation tools were streamlined to facilitate quick access to essential functions, addressing clinicians' workflow challenges during diagnostics. Contributions from ML researchers optimized model accuracy and interpretability, while HCI researchers ensured the UI adhered to user-centered design principles.

6.2. Formative user studies task

In most scenarios, breast cancer diagnosis is challenging due to diverse textures, the morphology of the lesions, and the density of the tissue, making accurate and consistent interpretation difficult for clinicians (Calisto et al., 2022b). Therefore, we focus on breast cancer diagnosis in imaging classification, a domain known for its high false positive rates (Seely and Alhassan, 2018). Our study aimed to address this challenge and the associated error rates by exploring the use of AI agents in multimodal medical imaging assessments.

In our study, clinicians diagnosed three patients with varying severities of breast abnormalities using different imaging modalities. They assessed the likelihood and location of malignancy by reading about five 2D images and one 3D volume per patient (Section 3.6). Finally, they classified the severity of each patient's condition using the BIRADS scale.

We compared conventional and assertiveness-based agents in assisting trained medical personnel in these diagnostic tasks. The conventional condition used the publicly available BreastScreening-AI framework (Calisto et al., 2022b), while we developed two additional conditions to customize AI outcomes for clinicians, one assertive and another non-assertive communication tones in a counterbalanced fashion. We conducted several trials with clinicians of different expertise levels, such as interns, juniors, middles, and seniors. The task was evaluating their interactions with the conventional, assertive, and non-assertive agents, providing insights into the effectiveness of different communication styles. Our goal was to identify the optimal communication strategies between the different expertise levels of clinicians.

6.3. Dataset

In this work, we used a comprehensive dataset comprising 338 patient cases obtained from HFF. Out of these, 289 cases were meticulously classified by the head of radiology, ensuring high accuracy and reliability in the dataset (Appendix A.3). The imaging data encompassed several modalities, including MG X-ray images captured in CranioCaudal (CC) and MedioLateral Oblique (MLO) views, UltraSound (US) images, and Dynamic Contrast-Enhanced MRI (DCE-MRI) slices. Each MRI case included multiple image slices capturing the lesion, significantly enhancing the dataset's richness and depth.

Table 2

Characteristics of participants for demographic groups with frequency and percentage. The main characteristics are gender, age, geographic location (clinical institution), and medical experience.

Demographic	Group	Frequency	Percentage
Gender	Female	36	69.23%
	Male	16	30.76%
Age	18–29	11	21.15%
	30–39	12	23.08%
	40–49	10	19.23%
	50–59	10	19.23%
	>59	9	17.31%
Geographic Location	Hospital 1	15	28.85%
	Hospital 2	12	23.08%
	Hospital 3	2	3.85%
	Hospital 4	9	17.31%
	Hospital 5	1	1.91%
	Hospital 6	1	1.91%
	Hospital 7	6	11.54%
	Hospital 8	1	1.91%
	Hospital 9	1	1.91%
	Hospital 10	1	1.91%
	Hospital 11	3	5.77%
Medical Experience	Interns	6	11.54%
	Juniors	17	32.69%
	Middles	11	21.15%
	Seniors	18	34.62%

Combining these modalities resulted in approximately 2890 images used to train and test the AI models, facilitating a comprehensive analysis and training process. The extensive dataset provided a robust foundation for developing and validating our intelligent agents. Consequently, the agents are well-equipped to assist in clinical decision-making, ensuring their reliability and effectiveness in varied clinical scenarios.

Pre-processing this dataset was critical to ensure consistency and quality across all images. We implemented several techniques, including resizing the images to a uniform dimension of 224×224 pixels, which is optimal for the DenseNet and ResNet models used in our study. Additionally, we normalized the images by subtracting the mean and dividing by the Standard Deviation (SD), a standard practice in DL to ensure the model learns effectively from the data. These pre-processing techniques were essential for addressing the diverse characteristics of the MG, US, and MRI images, which exhibit variations in intensity, resolution, and size. Standardizing the intensity levels across all modalities through normalization ensured consistent image quality, facilitating accurate analysis and model training.

6.4. Participants

We recruited 52 clinicians from various clinical settings (Tables 2 and 3), including public hospitals, cancer institutes, and private clinics, to participate in our study. This diverse participant pool offers valuable insights into AI communication strategies in various medical settings, enhancing understanding of AI tools' real-world applicability in healthcare.

From the demographic questionnaires, 55.77% of participants are expert clinicians, whereas 34.62% are seniors having more than 10 years of practical experience, and 21.15% are middle clinicians having more than 5 years but less than 10 years. Similarly, 44.23% of participants are novice clinicians, whereas 32.69% are juniors after taking the exam, having up to 5 years of clinical experience, and 11.54% are interns before the medical specialty exam. Each clinician was exposed to the three trials (*i.e.*, conventional, assertive, and non-assertive) in a counter-balanced manner.

The study's participant demographics offered a balanced representation of clinical expertise, ensuring diverse perspectives. Including intern, junior, middle, and senior clinicians allows comprehensive

analysis of how experience influences interaction with AI systems. Collaboration with multiple institutions enhances relevance and applicability, ensuring generalizable results. Rigorous selection and balanced exposure to different AI communication styles ensure robust, real-world insights.

6.5. Procedure

After providing informed consent, clinicians reported their demographic and professional backgrounds, including education (*e.g.*, radiology, surgery, nursing, or technical roles), areas of expertise, work sector, and medical experience (Section 6.4). They also shared details about their experience with medical imaging data. Clinicians then spent about three minutes familiarizing themselves with the user interface and shared functionalities of both AI agents.

At this stage, each participant interacted with the assistant, *accepting* or *rejecting* the system suggestion in the two different conditions: (a) conventional; and (b) assertiveness-based. The set of patients provided participants with 289 patients, while all patients must have at least one of the three available modalities. Each participant opened the set of three patients, chosen randomly, and examined it. As participants interacted with the system, they could use various functionalities and tools to support their diagnostic tasks (Table 1). These functionalities allowed clinicians to access relevant information, perform detailed analyses, and make informed decisions.

After each task, clinicians completed a brief questionnaire assessing their perceptions of the intelligent agents. The questionnaire measured trust across understanding, competence, and thoughtfulness, cognitive workload using NASA-TLX (Park et al., 2024), and usability with SUS (Wronikowska et al., 2021). It aimed to evaluate the agents' diagnostic utility, decision-making support, and potential adoption in practice. After completing all tasks, preferences for conventional or assertiveness-based agents were measured at varying levels of assertiveness.

6.6. Analysis

Our study explores how assertiveness-based agents affect clinicians' efficiency and diagnostic accuracy, focusing on communication styles and workflows. It compares traditional and assertiveness-based agents across scenarios. The study also details qualitative methodologies to gather clinicians' feedback, including interviews and focus groups. This feedback highlights preferences and needs in adopting AI tools.

For RQ1, we examined the impact of our assertiveness-based agent on clinicians' efficiency and accuracy in diagnosing patients using AI-suggested recommendations. A one-way ANOVA test (Sadeghi et al., 2022; Zhang et al., 2022) compared both AI agents regarding two outcome measures per clinician: (i) time taken for diagnosing each patient (H1.1.) and (ii) accuracy rates of false positives and false negatives (H1.2.). To evaluate efficacy differences between novice and expert clinicians during decision-making (H1.3.), we used a chi-squared test of independence (Reichherzer et al., 2021) to assess the relationship between clinician expertise and the assertiveness levels of the agents. Precision, Recall, and F1 Score were used to evaluate the diagnostic performance (Appendix A.5). Our dataset, validated through post-biopsy verification, allowed us to measure the patient's absolute ground truth.

For RQ2, we compared clinicians' perceptions of both conventional and assertiveness-based agents. We examined perceived preference (H2.1.) and trustworthiness (H2.2.) using the ANOVA test and statistical significance ($p < 0.05$). Scores for cognitive workload and usability (H2.3.) were compared between the two AI agents using statistical significance ($p < 0.05$) to compute confidence levels. A one-way ANOVA test assessed assertiveness-based levels in clinical arguments based on clinicians' professional experience. Specifically, we measured clinicians' perceived preferences regarding reliability and capability (H2.4.). The

Table 3

List of the eleven clinical institutions. All clinicians and institutions gave prior permission to use their data for research purposes under this study.

Number	Acronym	Full Name
Hospital 1	HFF	Hospital Fernando Fonseca
Hospital 2	IPOL	Instituto Português de Oncologia de Lisboa
Hospital 3	HSM	Hospital de Santa Maria
Hospital 4	IPOC	Instituto Português de Oncologia de Coimbra
Hospital 5	MMC	Madeira Medical Center
Hospital 6	SAMS	Serviços de Assistência Médico-Social do Sindicato dos Bancários do Sul e Ilhas
Hospital 7	HB	Hospital Barreiro Montijo
Hospital 8	HSA	Hospital de Santo António
Hospital 9	CF	Champalimaud Foundation
Hospital 10	CHTMAD	Centro Hospitalar Trás-os-Montes e Alto Douro
Hospital 11	CHLO	Centro Hospitalar Lisboa Ocidental

test ranged from “Totally Non-Assertive” (i.e., more suggestive) to “Totally Assertive” (i.e., imposing AI recommendations), evaluating the communication tone’s impact on both novice and expert clinicians.

Our analysis integrated open coding comments and feedback from focus groups, workshops, and interviews (Bieg et al., 2022; Shibuya et al., 2022). To effectively organize and analyze the clinicians’ responses, we employed affinity diagrams (Deutsch et al., 2019; Goldman et al., 2022), concentrating on their insights regarding workflow practices and the core functional concepts of the intelligent agents. These sessions promoted thorough discussions about the usability, interpretability, and effectiveness of AI agents in clinical decision-making. The primary aim was to identify and extract emerging themes from the discussions, enriching the context of our findings.

7. Results

We used a one-way ANOVA test with Python’s *scipy* library, focusing on medical professional experience. Our analysis, guided by literature recommendations (Casale et al., 2022), targeted statistically significant results² in areas such as time performance, accuracy, decision rates, preference choices, agreement comparisons, and perceptions of reliability and capability. These results highlight significant differences, emphasizing the impact of personalized AI communication on clinical workflows and decision-making efficacy. The findings detail the effects of assertiveness-based communication on novice clinicians, such as interns and juniors, as well as expert clinicians, including middle and senior levels.

7.1. RQ1: Impact on medical assessments

This section provides a consolidated overview of our key findings for **RQ1**, highlighting the impact of communication in AI-supported clinical decision-making. We discovered that clinicians’ time efficiency significantly benefited from the use of an assertiveness-based agent (Fig. 4), indicating the efficiency gains achievable with personalization AI communications in medical workflows. Significantly, this increased efficiency did not detract from the accuracy of diagnoses (Table 5).

The adaptability of the agent’s communication to align with different levels of clinician expertise (Table 4), points to the potential of AI systems in offering tailored experiences. The assertiveness-based approach enhanced decision-making. These findings suggest avenues for research on assertiveness-based intelligent agents, exploring their effects on engagement of the *end-users* and trust in diverse high-stakes domains.

The impact of an assertiveness-based agent on medical assessments was investigated in this section. We hypothesized that assertiveness-based communication with the intelligent agent would improve clinicians’ workflow and time performance (H1.1). The results confirmed

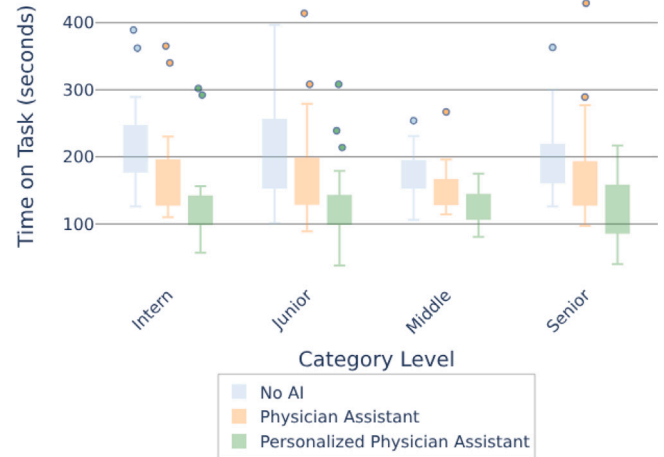


Fig. 4. The impact of personalized AI communication on clinician time performance. It shows the time taken for diagnosis by interns, juniors, middles, and seniors using the “No AI” tool, “Physician Assistant” (conventional AI), and “Personalized Physician Assistant” (assertiveness-based agents).

this hypothesis, as the time performance of clinicians improved significantly when using the assertiveness-based agent compared to the conventional agent (Fig. 4). The difference was statistically significant ($F = 11.32, p = 0.005 < 0.05$), indicating a large effect size ($r = 0.49$). These findings suggest customized communication in AI systems can enhance efficiency and optimize workflows.

Paired comparisons were conducted to examine the effects of assertive and non-assertive sub-conditions. Results revealed that assertive communication significantly enhanced efficiency for interns and juniors, with a 25.5% reduction in average decision time compared to non-assertive communication ($p = 0.03 < 0.05$). For middle and senior clinicians, however, non-assertive communication proved more effective, yielding a 14.2% improvement in decision accuracy ($p = 0.02 < 0.05$). The results highlight the varying impacts of communication tones by clinician expertise, emphasizing the need for adaptive strategies tailored to user experience levels.

Detailed analysis of diagnostic time across different clinician expertise levels revealed significant improvements (Fig. 4). Interns reduced their diagnostic time from 220.73 s (SD = 60.63) without AI to 130.40 s (SD = 35.14) with the personalized assistance. Juniors decreased their time from 213.92 s (SD = 67.34) without AI to 126.68 s (SD = 45.80) with the personalized assistance. Middle clinicians reduced their time from 194.08 s (SD = 48.53) without AI to 113.38 s (SD = 26.21) with the personalized assistance. Senior clinicians reduced their diagnostic time from 180.06 s (SD = 52.41) without AI to 112.52 s (SD = 34.10) with the personalized assistance. Overall, these findings demonstrate significant improvements in diagnostic time efficiency up to four times with personalized AI assistance compared to the current

² For more details about the achieved statistical results, the *sa-uta11-results* repository gives access to our exploratory analysis (<https://t.ly/e92CX>). Last accessed July 9, 2024.

Table 4

Percentage of clinicians' acceptance/rejection of AI recommendations between interns and juniors, as well as middle and seniors. Shows how often clinicians switched conclusions after interacting. "Overall Corrects" indicates correct acceptance/rejection. "Overall Mistakes" denotes incorrect acceptance/rejection. The highest percentage per column is marked in green, while the lowest percentage per column is marked in red.

Trials	Overall Corrects		Overall Mistakes	
	Interns & Juniors	Middles & Seniors	Interns & Juniors	Middles & Seniors
Conventional	69.70%	63.77%	30.30%	36.23%
Assertive	81.59%	65.76%	18.41%	34.25%
Non-Assertive	75.63%	66.41%	24.37%	33.59%

(No AI) medical setting.

Our findings underscore the effectiveness of assertiveness-based AI communication in significantly enhancing time performance across all clinician expertise levels, with intern and junior clinicians showing the most pronounced improvements. Intern and junior clinicians, including both interns and juniors, reduced their diagnostic time from an average of 171.35 s (SD = 42.86) with conventional assistance (PA — Physician Assistant) to 128.54 s (SD = 40.47) with personalized assistance (PPA — Personalized Physician Assistant). Middle and senior clinicians, comprising middle and senior clinicians, reduced their diagnostic time from an average of 146.94 s (SD = 35.31) with conventional assistance (PA) to 112.95 s (SD = 30.16) with personalized assistance (PPA). Results indicate that personalized assistance enhances diagnostic efficiency for intern and junior clinicians and provides significant time-saving benefits for middle and senior clinicians, highlighting the broad applicability and value of assertiveness-based AI communication across varying levels of clinical expertise.

In testing our hypothesis (H1.2), we aimed to determine if the assertiveness-based agent influenced clinicians' diagnostic accuracy (Appendix A.6). The results showed no significance ($F = 1.85, p = 0.37 > 0.05$) in accuracy between the assertiveness-based agent and the conventional agent (Tables 4 and 5). These findings suggest that the agent's explanations' assertiveness level does not negatively impact clinicians' ability to classify patients accurately. However, it is essential to note that a non-significant result does not necessarily imply equivalence in performance between the two conditions. Further investigation is needed to understand the nuanced effects of assertiveness-based communication in AI-mediated clinical decision-making.

This study examines how assertive vs. non-assertive communication styles impact the acceptance of AI recommendations among intern, junior, middle and senior clinicians (Table 4). Intern and junior clinicians tend to favor assertive communication (i.e., imposing the AI recommendations), achieving higher overall correct acceptance rates of 81.59% compared to 69.70% with conventional assistance and 75.63% with non-assertive agents. Conversely, expert clinicians show better accuracy with non-assertive agents (i.e., suggesting the AI recommendations), with correct acceptance rates of 66.41%, compared to 63.77% with conventional assistance and 65.76% with assertive agents. Our results indicate that the effectiveness of AI communication styles is contingent on the clinician's level of expertise, highlighting the need for tailored communication strategies to maximize diagnostic accuracy and efficacy.

Precision, Recall, and F1 Scores provide a comprehensive framework for evaluating diagnostic performance (Appendix A.5). We evaluated Precision, Recall, and F1 Scores for BIRADS classifications across AI conditions (Table 5). The PPA condition achieved the highest mean scores in Precision (0.896), Recall (0.882), and F1 Score (0.886), indicating superior diagnostic accuracy. The AI Model condition had the lowest Precision (0.864) and F1 Score (0.876), highlighting its ineffectiveness in balancing precision and recall. The No AI condition showed variability, with the highest Precision (0.92) and Recall (0.92) for BIRADS categories 1 and 5, but the lowest Recall (0.85) and F1 Score (0.88) for categories 1 and 2. The CPA condition demonstrated consistent performance, with a mean Precision of 0.876, Recall of 0.882, and F1 Score of 0.88. As *medium* severities (BIRADS 2 and 3)

Table 5

Precision, Recall, and F1 Scores for BIRADS classifications across four conditions: No AI, Conventional Physician Assistant (CPA), Personalized Physician Assistant (PPA), and AI Model. No AI refers to clinicians diagnosing without AI assistance. CPA provides BIRADS and accuracy without explanations. PPA offers personalized, adaptive recommendations with explanations. The AI Model serves as a comparison for the AI system's raw metrics. Metrics for BIRADS categories 1 through 5 are detailed, with green and red highlights indicating the highest and lowest values.

BIRADS	1	2	3	4	5	Total mean
No AI Precision	0.92	0.85	0.88	0.86	0.90	0.882
No AI Recall	0.85	0.82	0.90	0.88	0.92	0.874
No AI F1 Score	0.88	0.84	0.89	0.87	0.91	0.878
CPA Precision	0.87	0.84	0.89	0.87	0.91	0.876
CPA Recall	0.90	0.86	0.88	0.87	0.90	0.882
CPA F1 Score	0.89	0.85	0.89	0.87	0.90	0.880
PPA Precision	0.91	0.87	0.90	0.88	0.92	0.896
PPA Recall	0.88	0.85	0.89	0.88	0.91	0.882
PPA F1 Score	0.89	0.86	0.89	0.88	0.91	0.886
AI Model Precision	0.88	0.83	0.87	0.84	0.90	0.864
AI Model Recall	0.91	0.87	0.89	0.87	0.91	0.890
AI Model F1 Score	0.89	0.85	0.88	0.85	0.91	0.876

are known to be ambiguous (Section 3.5), lying between *low* (BIRADS 1) and *high* severities (BIRADS 4 and 5), the PPA condition simplified decision-making for borderline cases by enhancing accuracy and consistency.

To evaluate the accuracy variation influenced by the tone level (H1.3), we examined the impact of personalized explanations by customizing the agent's communication from interns to seniors (Table 4). The results revealed a significant association ($\chi^2 = 3.84, p = 0.001 < 0.05$) between the agent's communication tone and medical professional experience. Intern and junior clinicians had a higher chance of correctly classifying a patient with the assertive agent, while middle and senior clinicians had a slightly higher chance with the non-assertive agent.

The analysis compared decision rates between assertiveness-based and conventional agents. Clinicians made better decisions with assertiveness-based assistance, achieving higher acceptance and rejection rates. False positives were reduced by 27.59% for seniors, 18.84% for juniors, 20.10% for middles, and 25.22% for interns. False negatives were reduced by 4.60% for seniors, 2.90% for juniors, 3.36% for middles, and 2.52% for interns. Overall, customized AI systems can reduce false positives by 22.94% and false negatives by 3.35%, highlighting personalized communication's effectiveness in improving diagnostic accuracy across different expertise levels.

7.2. RQ2: Perception of clinicians

This section summarizes our investigation of question RQ2. The study explores clinicians' perceptions and interactions with AI in medical assessments, emphasizing preferences and trust in assertiveness-based communication and its implications for AI design in healthcare.

Clinician feedback provided insights into their experiences and attitudes towards assertiveness-based and conventional agents. Our statistical analysis revealed a significant preference ($F = 8.35, p = 0.001 <$

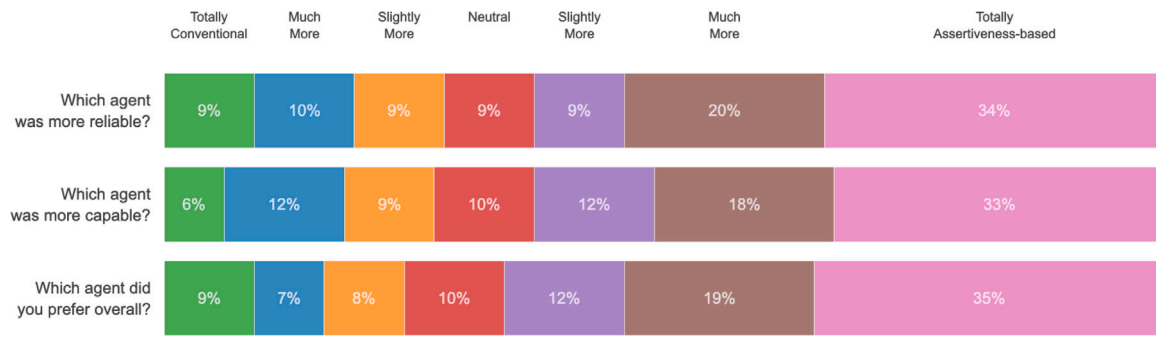


Fig. 5. Preference choices of clinicians when comparing both conventional and assertiveness-based agents within this study. Rates of clinicians are ranging from *Totally Conventional* to *Totally Assertiveness-based* on perceived *reliability*, *capability*, and *overall preference* of each agent.

Table 6

Comparison for the percentage of agreement between conventional and assertiveness-based agents. The questions are following the three *dimensions of trust* represented by perceived *understanding*; *competence*; and *thoughtfulness*.

Questions	Conventional	Assertiveness-based
Overall, I can trust in the agent recommendations.	86%	90%
I understand what the system is thinking.	91%	94%
The system seems competent.	82%	92%
The agent shows great thoughtfulness while dealing with the patient.	71%	75%

0.05) for the assertiveness-based agent, with 66% of participants favoring it (Fig. 5), as hypothesized in H2.1. Conversely, 24% preferred the conventional agent, and 10% did not express a specific choice. Indeed, the results highlight clinician receptiveness to assertiveness-based AI communication, suggesting its usefulness in medical decision-making.

Data showed that novice clinicians, such as interns and juniors, valued assertiveness-based communication, with 85% preferring it. They reported feeling more confident in their decisions, with 88% noting increased support. Assertiveness-based AI helps bridge clinical experience gaps by enhancing confidence and accuracy in decision-making.

Expert clinicians, including middle and senior levels, responded positively to assertiveness-based AI communication but with more nuanced preferences. About 78% appreciated its clear guidance, while 82% emphasized the importance of adapting communication to their expertise. They preferred a collaborative interaction where AI offers recommendations without undermining their judgment, with 85% favoring this approach.

In our study, we assessed clinicians' perceptions of trust, understanding, competence, and thoughtfulness towards conventional and assertiveness-based agents (Table 6). The statistical analysis revealed no significant difference in perceived trustworthiness between the two agents ($F = 19.47, p = 0.06 > 0.05$). Similarly, understanding levels did not show significant differences ($p = 0.14 > 0.05$). However, the assertiveness-based agent was rated significantly higher in terms of perceived competence ($p = 0.04 < 0.05$) and elicited higher thoughtfulness from clinicians ($p = 0.001 < 0.05$).

Our findings reveal a nuanced response to AI agents' communication styles, postulating careful interpretation. Although the non-significant differences in trust (Conventional: 86%, Assertiveness-based: 90%) and understanding (Conventional: 91%, Assertiveness-based: 94%) do not imply equivalence, the results partially support our hypothesis (H2.2.) concerning clinicians' preferences. The findings emphasize the need for personalized AI communication in high-stakes domains to enhance the end-user decision-making.

Workload and usability were analyzed between conventional and assertiveness-based agents using NASA-TLX and SUS measurements. Results indicated no significant differences in workload or usability between the two agent conditions ($p = 0.38 > 0.05$). These findings support our hypothesis (H2.3.) that assertiveness-based communication in AI agents would not affect clinicians' workload or usability. Perceived workload was reduced by 48%, while usability improved by 3.41%, suggesting that personalized AI recommendations can be integrated

without compromising ease of use or increasing cognitive burden.

Last, we examined how clinicians perceive the levels of assertiveness in communicating clinical arguments (Fig. 6). The results revealed significant differences in reliability ($F = 31.36, p = 0.0001 < 0.05$) and capability ($F = 18.17, p = 0.0003 < 0.05$) between intern, junior, middle, and senior clinicians. These findings support our hypothesis (H2.4.) that clinicians' preferences vary based on the assertiveness level of the agent, indicating differing perceptions of clinical arguments.

7.3. Qualitative insights

This section synthesizes the qualitative analysis of clinicians' perceptions and experiences with conventional and assertiveness-based agents. We adopted a participatory approach, analyzing data from 12 focus group sessions over four months. Together with our two ML and three HCI experts, sixteen participants with diverse expertise provided a broad perspective on interacting with AI agents.

Sessions offered insights into clinicians' challenges and expectations when using intelligent agents in decision-making. They highlighted the need to preserve image states for efficiency and accuracy and emphasized customizable communication tones to align AI interactions with their preferences and expertise. Quick access to external medical resources emerged as another key requirement, ensuring seamless integration of supplementary information into their workflows. Visual tools highlighting critical regions, such as lesion boundaries and severity indicators, were highly preferred, particularly using intuitive color-coded overlays.

Feedback underscored the importance of providing explanations that align with clinical reasoning. Clinicians valued detailed AI recommendations that included attributes such as lesion shape, margin, and density. Concerns frequently centered on cognitive overload caused by inefficient navigation tools and the absence of intuitive shortcuts. Such challenges highlighted the requirement for adaptive, user-centered interfaces. Effective design should accommodate varying levels of expertise, reduce potential errors, and streamline clinical workflows.

Our analysis showed a clear clinician preference for human-interpretable arguments over numerical AI outputs, with 92% favoring the assertiveness-based agent's personalized communication. This preference was linked to the agent's precise, meaningful explanations of AI recommendations. A junior clinician noted: "The assertiveness-based gave me a better understanding of the AI's reasoning, which helped me

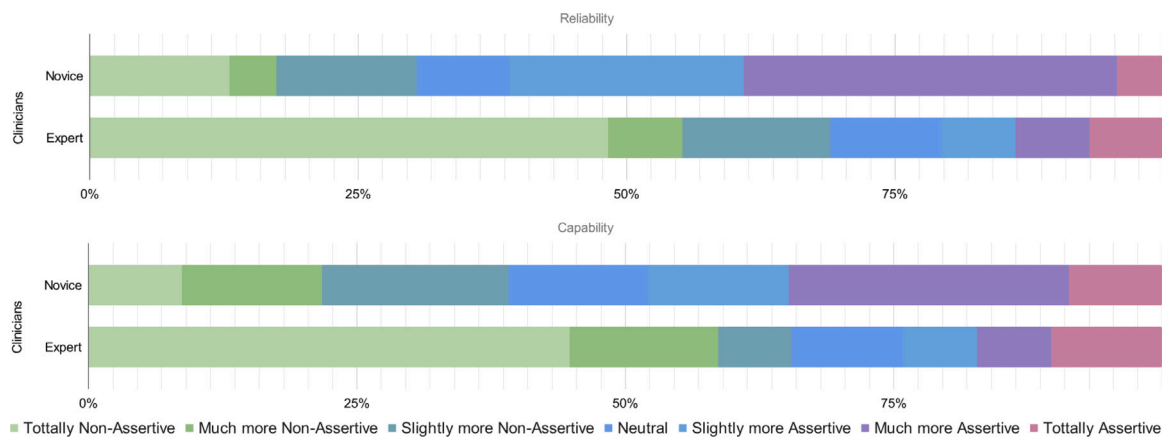


Fig. 6. Ratings between novice and expert clinicians for perceived reliability and capability. Clinicians rated each agent, ranging from *Totally Non-Assertive* to *Totally Assertive* communication.

make more informed decisions” (C6). Our findings also revealed how clinicians’ mental models of anthropomorphic intelligent agents are shaped by their experiences and expectations. One clinician remarked: “The first AI [assertiveness-based] was outstanding... but in the second AI [conventional] I was frustrated with the lack of communication in comparison to the first one” (C48). Expert clinicians used the tone of communication to gauge the reliability of AI recommendations, while novices focused on learning and patient comparisons.

These findings underscore the need for personalized and adaptable communication in AI systems, catering to clinicians’ diverse needs and expertise levels. They align with the following discussions, reinforcing the importance of user-centered design in developing AI systems for clinical decision-making (Zimmerman et al., 2022). Our research contributes to a broader understanding of how AI communication styles influence clinician decision-making, highlighting the need for technically proficient AI systems that are attuned to user needs and preferences.

7.4. Workflow impact

Personalized AI recommendations enhance diagnostic consistency in breast cancer diagnosis, leading to a measurable reduction in variability among clinicians’ decisions (Appendix A.7). Our study demonstrates that customized AI systems lower false positives by 22.94% and false negatives by 3.35% (Section 7.1). While the ground truth was based on expert radiologist consensus, AI helped clinicians achieve greater alignment in their diagnostic decisions. This consistency is crucial for evidence-based medicine, where reliable and repeatable interpretations form the foundation for accurate patient outcomes.

The observed reductions in diagnostic errors translate into significant workflow improvements. A decrease in false negatives, for example, increases the likelihood of timely treatments for critical conditions, enhancing patient prognosis by up to 5% in time-sensitive cases (Bray et al., 2024). Similarly, reducing false positives minimizes unnecessary follow-up procedures, alleviating patient anxiety and lowering healthcare costs. The 22.94% reduction in false positives equates to potential savings of up to 13.79% in diagnostic-related costs (Ong and Mandl, 2015), with a clinical setting serving 1,000 patients monthly potentially saving approximately \$229,400. These advancements underscore the broader value of integrating tailored AI systems into clinical workflows, offering tangible benefits for patient outcomes and healthcare efficiency.

AI integration also positively impacts clinician workload. Diagnostic efficiency improved by 1.38 times, significantly reducing the cognitive and physical demands of radiologists (Lauritzen et al., 2022). According to NASA-TLX measures, AI-supported workflows lowered workload perception by 48%, potentially reducing burnout rates from 20% to

40% (Alexander et al., 2022). Therefore, consistency in AI-supported diagnostics streamlines workflows, enhancing efficiency and reducing variability. Improved workflows support clinician satisfaction and maintain high-quality care. Integrating AI fosters sustainable healthcare practices while addressing clinician well-being and mitigating workload stress.

While accuracy remains a critical goal, our findings suggest that personalized AI systems improve the interpretative consistency among clinicians of varying expertise. This consistency reduces diagnostic variability, indirectly supporting better patient outcomes and providing a reliable foundation for future AI-assisted diagnostic tools. Further investigations are required to explore how enhanced consistency translates to long-term improvements in diagnostic accuracy.

8. Discussion

In this study, we examined how personalized communication can enhance the effectiveness of intelligent agents in clinical decision-making within medical imaging. A within-subject design was employed, allowing clinicians to interact with both conventional and assertiveness-based agents. The assertiveness-based condition encompassed both assertive and non-assertive communication styles, grouped conceptually to evaluate tone variability within a unified framework. This approach provided a nuanced understanding of how varying communication strategies influence clinician efficiency and diagnostic accuracy. Findings demonstrated that assertiveness-based agents significantly improved workflow efficiency without compromising diagnostic accuracy, offering clear benefits in reducing decision-making time and supporting clinical performance.

Efficient decision-making is central in medical imaging during radiological diagnosis (van der Stappen and Funk, 2021; Su et al., 2022). Our results indicate marked improvements in time efficiency, vital in high-stakes environments where timely decisions significantly affect outcomes. These gains did not compromise accuracy, highlighting the value of assertiveness-based communication in clinical settings. The study emphasizes the importance of tailored communication for addressing clinicians’ needs and expertise.

Personalized AI communication impacts clinician performance differently based on their experience. Intern and junior clinicians responded positively to assertive communication, gaining confidence and guidance for accurate diagnoses. While appreciating the clarity of assertive agents, middle and senior clinicians preferred a more collaborative approach that respects their judgment. This differential response underscores the need for intelligent systems to adapt communication strategies based on user experience and expertise. Integrating intelligent agents in clinical practice can reduce workload, address burnout, and improve job satisfaction. Given the increasing demands

on healthcare professionals and the prevalence of burnout (Alexander et al., 2022; Lauritzen et al., 2022), assertiveness-based communication can enhance AI systems' perceived reliability and competence.

Clinicians reported higher trust and perceived competence with anthropomorphic intelligent agents that adapt to user demographic characteristics. Adapting to these characteristics is essential for AI adoption in clinical settings (Calisto et al., 2022a; Sivaraman et al., 2023; Wang et al., 2021). Trust in AI systems is crucial for their integration into daily practice. Tailored communication can significantly build this trust.

8.1. Clinical workflow efficiency

Understanding AI communication's impact on clinical workflow efficiency is vital (Wang et al., 2021). Tailored AI systems must align with clinicians' varying expertise levels (Zhang et al., 2022). Our study examines how assertiveness-based AI communication affects workflow efficiency, particularly for interns, juniors, middles, and seniors. Personalized AI communication bridges the gap between novice and experienced clinicians, ensuring enhanced support for all. Results show that assertiveness-based AI communication significantly improves workflow efficiency (RQ1), especially for intern and junior clinicians.

The significant decrease in diagnostic time for less experienced clinicians underscores the effectiveness of clear and directive AI guidance. Our findings support the hypothesis that assertiveness-based AI communication improves clinician efficiency (H1.1). Notably, the increase in efficiency does not compromise diagnostic accuracy.

The improvement in diagnostic accuracy remained consistent across different levels of clinician experience (H1.2). The findings show that assertiveness-based communication did not negatively impact diagnostic outcomes. Consistent accuracy across varying expertise levels highlights the robustness of personalized AI support. This reliability ensures AI systems maintain high standards of clinical accuracy, regardless of the clinician's experience. Consequently, personalized AI communication enhances efficiency and sustains the quality of diagnostic decisions, vital for effective patient care (Rodríguez-Ruiz et al., 2019).

Our enhanced performance of assertiveness-based agents over conventional AI systems stems from the richer informational content in assertiveness-based conditions. While some might view the added depth as potentially overwhelming, it mirrors real-world clinical interactions, where communication style and the level of detail are closely intertwined. Notably, the marked gains in efficiency and perceived support, particularly among less experienced clinicians, highlight the dual importance of communication style and content. Together, these elements significantly enhance the effectiveness of AI-clinician interactions, emphasizing their role in improving clinical decision-making and workflow integration.

In this work, the reduction in diagnostic time was less pronounced for middle and senior clinicians. Yet, it underscores the utility of personalized AI support. Despite their experience, these professionals benefit from tailored AI communication. Results indicate that while efficiency improvement varies, customized AI communication continuously enhances the clinical workflow by informing clinicians of the regions of interest for each lesion. This highlights the adaptability power of AI systems in providing meaningful support across different expertise levels.

The findings also reveal an interesting aspect of accuracy variation influenced by the communication tone level of the assertiveness-based agent (H1.3). Interns and juniors showed more significant improvements with assertive communication, while middle and senior clinicians responded better to a more collaborative and suggestive tone. This differentiation underscores the need for AI systems to dynamically adjust their communication strategies based on the clinician's level of expertise.

Overall, the study underscores the importance of designing AI systems that address the diverse needs of clinicians at various career stages. Tailored communication enhances workflow efficiency and supports precise decision-making, accommodating the unique requirements of clinicians with varying levels of expertise. Adaptable AI systems effectively bridge the gap between novice and experienced clinicians, ensuring relevance and usability across clinical practice. Such flexibility is essential for maintaining the utility and impact of AI tools in dynamic and high-stakes healthcare environments.

8.2. Trust and perception of AI systems

Trust in AI systems is essential for practical integration into clinical practice (RQ2). The study revealed that assertiveness-based AI communication enhances perceived competence and trustworthiness among clinicians. Interns and juniors who reported higher trust levels with assertive AI felt more supported and confident in their diagnostic decisions.

Middle and senior clinicians reacted favorably to assertiveness-based AI, though their preferences were more complex. Experienced clinicians valued the clear directives of personalized AI but emphasized the need for communication styles that respect their expertise. Adaptive communication ensures the AI system enhances decision-making without detracting from professional judgment. The finding supports our hypothesis regarding clinicians' preferences for assertiveness-based conditions (H2.1).

Communication styles that adapt to the clinician's experience align with our hypothesis on trustworthiness (H2.2). Experienced clinicians indicated that trust in AI systems could be significantly enhanced when AI communication acknowledges and integrates their professional expertise and judgment. Such adaptation fosters more collaborative interaction, ultimately improving the efficacy of AI-assisted clinical decision-making.

Clinicians' workload and usability perceptions were examined to understand the impact of tailored AI communication on daily practice (H2.3). Results showed no significant increase in workload or decrease in usability with assertiveness-based AI communication. Personalized explanations and tailored highlights do not add to clinicians' cognitive burden, ensuring the AI system remains practical and user-friendly. Maintaining a balanced workload is critical for clinician satisfaction, confirming that tailored AI can be integrated into clinical workflows without negatively impacting usability.

Perceptions of reliability and capability varied significantly between different levels of clinical expertise, highlighting the importance of adaptive communication strategies (H2.4). Interns and juniors showed higher reliability and capability ratings for assertive communication, while middle and senior clinicians preferred a more suggestive approach. The need for AI systems to dynamically adjust their communication styles to match the clinician's expertise is clear. Adapting the communication tone based on the clinician's experience ensures the AI system is perceived as a reliable and capable assistant, fostering trust and enhancing decision-making.

8.3. Toward personalized AI communication in HCI

Personalized AI communication tackles a fundamental challenge in HCI: **creating systems that adapt to the diverse needs of their users.** The study highlights the advantages of assertiveness-based communication in clinical workflows while pointing to a broader principle, *i.e.*, the need for user-centered design to foster trust, improve efficiency, and support accurate decision-making. The findings underscore that generic, *one-size-fits-all* communication strategies fail to meet the complex requirements of heterogeneous user groups. Instead, personalization tailored to individual users or specific subgroups enhances both usability and system effectiveness.

Interns and junior clinicians benefitted most from assertive guidance, providing confidence and decision-making structure. Conversely, middle and senior clinicians preferred collaborative and suggestive tones, which aligned with their expertise and autonomy. Our findings suggest that adaptive AI systems, capable of tailoring communication strategies in real-time, are essential for achieving optimal outcomes across varied expertise levels.

The implications extend far beyond healthcare. Personalized AI communication has the potential to transform user interactions in education by adjusting instructional styles based on students' proficiency and learning preferences. In customer service, such systems could tailor responses to suit customer temperaments, enhancing satisfaction and problem resolution. Similarly, adaptive communication can bridge gaps between novice and experienced employees in workplace decision support, fostering more efficient collaboration and better outcomes.

Advancing personalized AI communication requires further exploration of dynamic adaptation mechanisms. Techniques such as real-time user modeling, which analyzes behavior and preferences, could enable AI systems to fine-tune responses instantly. Additionally, integrating XAI capabilities can help users understand the rationale behind adaptive communication, further strengthening trust and usability.

To summarize, our study underscores the need for multidisciplinary approaches in designing adaptive AI systems. Combining HCI, cognitive psychology, and machine learning insights can create robust solutions that address user diversity while maintaining system transparency and reliability. Personalization is not just a feature but a necessity for AI systems aiming to achieve optimal outcomes across domains where human-AI collaboration is essential.

8.4. Design recommendations for adaptive AI communication

Personalized AI communication systems must evolve to meet diverse user needs dynamically. Two key areas require attention to enhance the design and application of such systems. First, incorporating a neutral communication condition can help disentangle the roles of assertiveness and informational content in improving decision-making. Isolating these factors would clarify whether benefits arise from communication tone or the additional guidance assertive communication provides. Exploring hybrid designs, where conventional AI integrates varying levels of explanation, offers an opportunity to identify the ideal balance between delivering directive support and minimizing cognitive load.

Second, the practical implementation of adaptive AI communication depends on robust user modeling. As used in this study, predefined subgroups provide valuable insights but fall short in real-world scenarios that demand dynamic adaptation. Systems must infer user expertise in real time to personalize interactions effectively. Self-perceived expertise does not always align with actual performance, creating challenges for AI systems that rely exclusively on user self-assessments. Integrating explicit user inputs, such as self-reported expertise levels, with implicit metrics like task performance and interaction patterns can address this issue. Such a combination enables the development of adaptive models to assess UX in real-time, tailoring system responses effectively.

ML techniques enable systems to adapt dynamically to user behaviors and preferences, advancing user modeling. Behavioral analytics and contextual data can inform systems that adjust communication strategies dynamically, ensuring responsiveness and relevance. These enhancements could extend beyond clinical settings to diverse domains such as education, customer service, and smart home technologies. Expanding research into these non-clinical contexts will help establish the generalizability of personalized AI communication strategies and refine their application in varied HCI scenarios.

Prioritizing thoughtful design considerations can transform AI systems into adaptive and intuitive tools that meet diverse user needs. Such systems enhance trust, improve efficiency, and support decision-making across varied expertise levels. Advancements in personalization and adaptability hold the potential to create AI systems that are both equitable and effective, making them valuable in specialized domains like healthcare and everyday applications.

9. Challenges and opportunities

This study provides insights into designing assertiveness-based AI communication systems for clinical workflows. The findings highlight the potential of personalized communication to enhance efficiency, trust, and decision accuracy, while several challenges require attention. Addressing these is vital for ensuring the robustness and practicality of such systems in real-world settings. The study also suggests avenues for exploring advanced personalization strategies in dynamic, high-stakes environments like healthcare. This section discusses the study's limitations and directions for future research.

9.1. Limitations

The study employed a within-subject experimental design to examine clinicians' interactions with assertiveness-based agents in breast cancer diagnosis, specifically focusing on the Assertiveness-based BreastScreening-AI framework. The design allowed all participants to experience conventional and assertiveness-based communication conditions, ensuring a comprehensive evaluation. Within the assertiveness-based condition, non-assertive and assertive communication styles were grouped to explore tone variability while maintaining a consistent conceptual framework.

The study did not include a 'neutral' condition where explanations are provided without assertiveness. This omission limits the ability to disentangle the effects of assertiveness from the additional informational content provided in assertiveness-based conditions. However, the design reflects practical clinical scenarios where assertive communication is often coupled with tailored informational content. Future research could explore a controlled comparison with neutral conditions to isolate these effects further.

Challenges arose due to the limited availability of clinicians and the study's remote nature, making it difficult to control every aspect of the experimental procedure. Variations in participant task completion times may have influenced their initial experiences with the assertiveness-based agent. Despite these limitations, the study offers valuable insights into how communication styles can be adapted to meet the needs of clinicians at varying expertise levels.

An additional challenge arises from the evolving legal framework concerning AI liability in medical settings. AI poses complex questions that are not fully addressed by current laws, making determining liability for errors or harm difficult. Furthermore, integrating AI into clinical workflows presents challenges in assigning responsibility for diagnostic decisions, mainly when clinicians act on recommendations provided by AI systems.

Current legal standards may not cover scenarios where AI contributes to diagnostic errors, leading to ambiguity in accountability. Policymakers need to develop frameworks and guidelines to address these issues, ensuring clarity in liability and fostering trust in AI technologies. Ongoing legal research and advancements are essential for a supportive environment for AI adoption in clinical practice.

9.2. Future directions

Insights from this research contribute to healthcare, HCI, decision support, and AI communication fields. Future research should explore how different communication styles impact clinical decision-making (Appendix A.8). Further investigation is needed to determine how intelligent agents can be optimized to support clinicians in various medical contexts.

Prior research has shown the potential to predict clinicians' trust in AI recommendations using raw medical data (Raghu et al., 2019). Our study utilized an assertiveness-based agent with AI outputs curated to reflect typical clinical scenarios. The next step involves training DL models to generate personalized explanations, serving as human-interpretable arguments. This enhances transparency and helps clinicians understand and accept AI recommendations.

Including a neutral condition in future research could clarify the individual effects of assertiveness-based communication and informational content in AI interactions. Separating these variables would enable a more precise analysis of their independent roles in influencing clinician decision-making. Such an approach could deepen our understanding of how communication style and content contribute to the effectiveness of AI-supported clinical workflows.

Another promising area of investigation involves exploring hybrid communication conditions in AI systems. Conventional AI could be augmented with explanations of varying depth, creating more nuanced and adaptable interactions. Examining such hybrid approaches would allow researchers to analyze the interplay between communication strategies and their impact on decision-making. Insights from these studies could reveal how different communication styles influence clinicians at various levels of expertise, ultimately guiding the design of more effective AI systems tailored to diverse professional needs.

Future work should examine the distinct effects of the assertiveness-based agent's primary features, explanations, and tone in different conditions. Additionally, studies should explore how different communication styles impact clinical decision-making. Understanding the effects of varied demographic characteristics among clinicians, such as those from medical institutions (public hospitals, private clinics, cancer centers) or specialties (family physicians, breast surgeons), could reveal further enhancements in AI-clinician interactions. A multidisciplinary approach can improve personalized communication between agents and clinicians, optimizing AI support across diverse medical contexts. Further investigation is needed to understand the nuanced effects of assertiveness-based communication in AI-mediated clinical decision-making.

10. Conclusion

This study offers a novel perspective on personalizing and customizing the explanations of intelligent agents for clinicians. In an experimental study with 52 clinicians, we compared the performance of a conventional agent with an assertiveness-based agent. Our findings reveal that adaptive communication tones and detailed patient case explanations significantly enhance workflow efficiency and diagnostic accuracy.

The assertiveness-based agent led to a considerable drop in diagnostic time per patient, reducing it by 27.54% for interns and junior clinicians and by 27.01% for middle and senior clinicians. Overall, clinicians were almost 1.38 times faster, meaning they completed their tasks about 27% quicker and made about 26% fewer diagnostic errors with the assertiveness-based agent compared to the conventional AI condition. These results underscore the importance of clear and directive AI guidance for less experienced clinicians and the continued relevance of personalized communication for highly trained professionals.

The assertiveness-based agent significantly reduced cognitive load, achieving up to a 48% decrease. This reduction is evidenced by shorter time-on-task and enhanced decision-making efficiency, all without sacrificing diagnostic accuracy. The consistent accuracy across different expertise levels underscores the robustness of personalized AI support, maintaining high clinical standards irrespective of the clinician's experience.

Clinicians reported higher trust and perceived competence with the assertiveness-based agent due to its tailored communication and detailed explanations. The study highlights the need for AI systems to adjust communication strategies based on the clinician's expertise, fostering a more collaborative and supportive interaction.

Our work has significant implications for designing AI systems in medical and other fields that require personalized human-AI interactions. Future research should explore the effects of different communication styles and refine intelligent agents to optimize support for clinicians in diverse medical contexts. Examining the long-term impact of assertiveness-based AI communication on clinician performance and patient outcomes is crucial for advancing AI integration in healthcare.

CRedit authorship contribution statement

Francisco Maria Calisto: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **João Maria Abrantes:** Validation, Data curation. **Carlos Santiago:** Writing – review & editing, Software, Methodology. **Nuno J. Nunes:** Writing – review & editing, Validation, Supervision, Conceptualization. **Jacinto C. Nascimento:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Francisco Maria Calisto reports financial support, administrative support, article publishing charges, statistical analysis, travel, and writing assistance were provided by Foundation for Science and Technology. Francisco Maria Calisto reports a relationship with University of Lisbon that includes: funding grants, non-financial support, speaking and lecture fees, and travel reimbursement. Francisco Maria Calisto has patent Computational Method and System for Improved Identification of Breast Lesions issued to Instituto Superior Técnico. Francisco Maria Calisto has patent Machine Learning System for Diagnosing Breast Lesions, Adaptable to the Clinician and Based on Assertiveness pending to Instituto Superior Técnico. The authors are external advisors of the BreastScreening-AI project (SensiPerception, LDA), a spinoff of Instituto Superior Técnico, University of Lisbon. In their advisory capacity, they contribute to the strategic direction and research of the BreastScreening-AI project, which focuses on enhancing breast cancer screening through advanced AI technologies. Their roles include providing expert guidance on algorithm development, clinical validation, and the integration of AI tools into clinical workflows. It is important to note that the project did not receive any funding. The authors declare no competing financial interests or personal relationships that could have influenced the work reported in this paper. These declarations are made to ensure transparency and uphold the integrity of the submitted work. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting details

Additional details on the application of AI models in our UI, severity classification during patient diagnosis, and patient selection approach are provided in this appendix. The current system is reviewed, focusing on performance recognition, thresholds, and patient curation strategies. Future steps for explanations and tone are outlined. The following sections elucidate these aspects to enhance understanding of our work.

A.1. Severity classification

BIRADS stands for “Breast Imaging Reporting and Data System” and is a system used to standardize how radiologists report the findings of mammograms and other imaging exams of the breast (Spak et al., 2017). The BIRADS provides a standardized method for reporting the results of breast imaging exams (Section 3.1), which can help to ensure that the information is accurate and consistent.

The BIRADS is helpful as an input for AI models designed to assist with diagnosing breast cancer (Section 3.2), as it provides a standardized way of representing the findings of imaging exams (Maicas et al., 2019). Using the BIRADS system as an input for AI models may improve the accuracy and reliability of the model’s predictions and help prevent bias in the results. The BIRADS system uses a scale from 0 to 6 for categorizing the findings of breast imaging exams (Section 3.3). However, in our study, we just considered the scale from 1 to 5, as the 0 means that the case is inconclusive, where we need to acquire more images, and 6 means we already have biopsy confirmation by the previously known lesion.

Here is a brief overview of each category on the BIRADS scale:

1. **Negative:** The exam showed no abnormalities, and the patient’s breast tissue appears normal.
2. **Benign Finding:** The exam showed a benign (non-cancerous) abnormality in the breast tissue.
3. **Probably Benign:** The exam showed an abnormality likely benign, but further testing may be needed to confirm this.
4. **Suspicious Abnormality:** The exam showed an abnormality that is suspicious for cancer, and further testing, such as a biopsy, is needed to determine if it is cancerous.
5. **Highly Suggestive of Cancer:** The exam showed an abnormality that is highly suggestive of cancer, and a biopsy is recommended to confirm the diagnosis.

In this work, we considered a *low* severity as BIRADS of 1, *medium* severities as BIRADS of 2 or 3, while *high* severities are considered as BIRADS of 4 or 5. Yet, choosing between each score remains ambiguous (Section 3.4), even for human radiologists. It is important to note that the BIRADS scale is only a tool for reporting the results of breast imaging exams and does not provide a definitive diagnosis of cancer. A biopsy is usually needed to confirm a cancer diagnosis.

A.2. Ambiguity in BIRADS categories

BIRADS categories are essential for standardizing breast imaging reporting and facilitating clinical decision-making (Section 3.5). The BIRADS helps ensure consistency and clarity in reporting, which is crucial for effective patient management. However, specific categories and borderline cases pose significant challenges due to their inherent ambiguity.

BIRADS 2: Benign Findings — This category includes definitively benign findings with a 0% risk of malignancy. Despite its apparent clarity, differentiating between BIRADS 2 and categories with slight suspicion can be challenging, mainly when subtle imaging features are present that might suggest otherwise.

BIRADS 3: Probably Benign — Assigned to findings with a high probability of benign (less than 2% risk of malignancy), BIRADS 3 depends on short-interval follow-up imaging to ensure stability. The term “probably benign” introduces uncertainty, as it encompasses a range of non-malignant but atypical features, potentially leading to variability in diagnostic interpretation and patient anxiety.

BIRADS 4: Suspicious Abnormality — This category, indicating a suspicious abnormality with a 2%–95% chance of malignancy, is subdivided into 4A (low suspicion, 2%–10%), 4B (moderate suspicion, 10%–50%), and 4C (high suspicion, 50%–95%). The broad risk range within BIRADS 4 contributes to its ambiguity, necessitating biopsy for definitive diagnosis. Distinguishing between subcategories accurately is

crucial but often challenging, requiring careful assessment of imaging characteristics. In this work, we used the single BIRADS 4 score for the simplicity of the decision-making process.

Borderline Cases — Ambiguity is particularly pronounced in borderline cases: (i) between “no findings” (BIRADS 1) and “benign” (BIRADS 2 and 3), where even minor imaging anomalies must be carefully evaluated to avoid unnecessary follow-ups or missing a potential issue; and (ii) between “probably benign” (BIRADS 2 and 3) and “suspicious abnormality” (BIRADS 4), where the decision to recommend a biopsy or additional imaging can significantly impact patient management and outcomes.

Borderline cases challenge even experienced radiologists due to their subtle features. Addressing the ambiguity in BIRADS 2, 3, and 4 while managing borderline cases is crucial for enhancing diagnostic consistency and accuracy. Advanced diagnostic tools, such as personalized AI systems, offer support with tailored recommendations to address such complexities.

Personalized AI systems can adapt their suggestions to align with the clinician’s expertise level, whether they are novices or seasoned professionals. Such adaptability not only enhances diagnostic precision but also promotes confidence in decision-making. To address these challenging scenarios effectively, personalized AI systems contribute to delivering high-quality, patient-centered care.

Future research should refine AI algorithms to assist radiologists with ambiguous and borderline cases. Enhancing these scenarios will improve breast cancer diagnoses and patient outcomes. Advanced algorithms are crucial in reducing diagnostic errors and providing definitive guidance in clinical practice. Incorporating clinician feedback ensures that AI systems align with clinical workflows. Collaboration between researchers and medical professionals is essential to design systems that integrate seamlessly into clinical environments.

A.3. Patient & modality selection

Mammography (MG), Ultrasound (US), and Magnetic Resonance Imaging (MRI) volumes are critical modalities for diagnosing and managing breast cancer (Calisto et al., 2020). Each modality offers unique advantages and complements the others, providing a comprehensive view of breast tissue. Women typically undergo routine screening mammograms, which are X-ray images of the breast (Lauritzen et al., 2022). However, mammograms alone can miss up to 27% of cancers (Bray et al., 2024), particularly in women with dense breast tissue.

When mammograms reveal suspicious findings, additional tests such as US images and MRI volumes are often necessary (Ahn et al., 2023). US images help differentiate solids from cystic masses and provide real-time imaging, crucial for guiding biopsies and interventional procedures. Their ability to detect lesions in dense breast tissue, where mammograms may be less effective, makes them essential in breast imaging. MRI offers superior soft-tissue contrast and is highly sensitive in detecting breast cancer, especially in high-risk patients and those with dense breast tissue. It is often used for screening high-risk women, evaluating cancer extent, and assessing treatment response.

Despite these modalities, the diagnostic process can be slow, stressful, and sometimes inaccurate, with errors ranging from 30% to 50% (Alexander et al., 2022; Sharma et al., 2022). There is a pressing need for a faster, more reliable solution that combines data from these three modalities (Calisto et al., 2021). After a mammogram identifies a region of interest, a US is performed to pinpoint the location. MRI then helps map the cancer’s size, position, and spread, which is critical for accurately targeting the mass for biopsy.

Our AI models are trained using data from MG, US, and MRI modalities, offering a more comprehensive analysis. This approach ensures that AI models do not rely solely on mammograms, which lack the detailed information provided by US and MRI. These modalities offer precise localization and volumetric data, enhancing diagnostic accuracy. Integrating these three imaging techniques into AI-assisted

diagnostic tools can significantly improve the quality of care and outcomes for patients.

We used 338 patient cases from the HFF clinical institution (Table 3). Of these, 289 were classified by the head of radiology and curated by us. Each patient has several images, including four 2D X-ray MGs of the left and right breasts in CranioCaudal (CC) and MedioLateral Oblique (MLO) views (CC-L, CC-R, MLO-L, MLO-R). Additionally, each patient has one or two 2D US images and about five Dynamic Contrast-Enhanced MRI (DCE-MRI) volumes. The MRI volumes are 3D images consisting of numerous slices per patient, capturing detailed images of the lesion.

From the 289 classified cases, we selected a total of 35 patients to be classified by our AI models. Because we aim to test the three trials (*i.e.*, conventional vs. non-assertive vs. assertive), plus the four groups of medical professional experience (*i.e.*, intern, junior, middle, and senior clinicians) we computed at least $2^5 = 32$ the number of patients. Hence, 35 patients were selected, curated, and manually annotated by another radiologist to cover that magnitude of patients. This classification corresponds to assigning a BIRADS value for each modality image of the exam.

The selection of cases was guided by the need to explore a broad spectrum of diagnostic challenges, including textbook and uncertain cases, reflecting the diversity encountered in clinical practice. Cases were sourced from a dataset of 338 patients curated by the head of radiology, ensuring their clinical relevance and alignment with study objectives. Emphasis was placed on including cases with varying degrees of ambiguity, such as borderline BIRADS categories (*e.g.*, between 3 and 4), which often require nuanced interpretation and decision-making. This diverse selection allowed for a comprehensive evaluation of the AI system's adaptability and effectiveness across varying levels of diagnostic complexity.

Uncertain cases were deliberately included to evaluate the AI system's performance under conditions where clinicians typically face the most significant diagnostic challenges. This approach assessed how different communication styles influence decision-making in high-stakes and ambiguous scenarios. For example, clinicians reviewed cases with subtle imaging anomalies or conflicting findings across modalities, which are more likely to benefit from personalized AI assistance. Including such cases offered insights into the system's ability to improve diagnostic confidence and accuracy in complex environments.

The study ensured a comprehensive assessment of the AI agents' effectiveness by including cases that represented a mix of straightforward and complex diagnostic scenarios. This setup allowed us to explore how assertiveness-based communication impacts clinicians' trust, efficiency, and accuracy, particularly in scenarios where AI support is most needed. Integrating such cases also provided valuable insights into the system's ability to simplify decision-making and enhance confidence in ambiguous situations.

A.4. Existing AI models

The proposed approach introduces a flexible and dynamic system that addresses the limitations of the traditional framework (Calisto et al., 2022b). It offers a robust and scalable solution to current diagnostic challenges (Section 3.6). The methodology enhances diagnostic accuracy by employing independent analyses for each imaging modality.

The AI models employed in this study operate independently for each imaging modality (*e.g.*, MG, US, MRI), without fusing their predictions. Each modality is assigned its own ground-truth score (Appendix A.6), representing the precise diagnosis for a patient (Appendix A.7). This separation is crucial, as different imaging modalities offer unique diagnostic insights into breast tissue, potentially leading to varying diagnoses and BI-RADS scores.

For 2D imaging data, such as MG and US images, the DenseNet model (Huang et al., 2019) was employed. DenseNet's densely connected convolutional layers facilitate efficient feature propagation and mitigate the vanishing gradient problem, making it highly effective for 2D image analysis.

Conversely, the 3D ResNet model (Aldoj et al., 2020) was employed for 3D imaging data, such as MRI volumes. ResNet's use of residual learning addresses the challenges of increased network depth, ensuring accurate volumetric data analysis. The lesion score, indicating the likelihood of malignancy in specific breast tissue regions, is a critical component of the BI-RADS scoring system, standardizing radiological reporting.

A key contribution of this study is modality-specific analysis, which enables tailored diagnostic insights and reduces the risk of errors inherent in single-modality systems. This approach achieves higher accuracy and reliability, leveraging the unique strengths of each imaging modality. The methodology underscores the importance of independent analysis for each modality, improving diagnostic precision and effectiveness, thus significantly advancing breast cancer diagnosis.

A.5. Performance evaluation

Deploying AI models in breast cancer diagnostics requires rigorous evaluation to ensure reliability and clinical utility. Accurate diagnosis impacts patient outcomes, influencing treatment decisions and prognoses. Misdiagnoses can lead to unnecessary interventions or missed treatments. Precision, Recall and F1 Score validate AI model effectiveness, providing a comprehensive assessment for improvements.

Performance evaluation in this study involves the metrics of Precision, Recall, and F1 Score, which are crucial for assessing the effectiveness of the AI models in diagnosing breast cancer. Precision measures the accuracy of the optimistic predictions, indicating the proportion of true positive predictions among all optimistic predictions made by the model. Recall evaluates the model's ability to identify positive cases, reflecting its sensitivity. The F1 Score, as the harmonic mean of Precision and Recall, provides a balanced measure that accounts for both false positives and false negatives, ensuring a comprehensive evaluation of the model's performance. Together, these metrics offer a detailed understanding of the AI models' diagnostic capabilities and areas for improvement.

Beyond their definitions, these metrics have important real-world implications. Optimizing Recall is vital in early cancer detection to prevent missed cases, while Precision reduces patient anxiety and resource strain from false positives. Precision and Recall together curves reveal model performance across thresholds, highlighting trade-offs aligned with clinical priorities. Domain-specific evaluations ensure robust AI systems support both the practical and emotional aspects of clinical decision-making, fostering trust and adoption.

A detailed understanding of these metrics allows for targeted refinements in the AI system, ensuring it effectively supports clinicians in delivering accurate and timely diagnoses. The models can be fine-tuned by addressing specific weaknesses revealed through these evaluations to achieve higher reliability and greater trustworthiness in real-world clinical applications.

Precision measures the proportion of correctly predicted positive instances out of all cases predicted as positive. It answers the question: **"Of all the instances that the model predicted as positive, how many were actually positive?"**

It is defined as follows:

$$\text{Precision} = \frac{TP}{TP + FP}$$

where TP represents true positives and FP represents false positives. A higher precision indicates fewer false positives, which is critical in minimizing unnecessary biopsies.

Recall (also known as sensitivity or true positive rate) measures the proportion of correctly predicted positive instances out of all actual

positive instances. It answers the question: “Of all the instances that were actually positive, how many did the model correctly identify?”

The recall is calculated as:

$$\text{Recall} = \frac{TP}{TP + FN}$$

where FN represents false negatives. High recall ensures that most cases of breast cancer are detected, reducing the likelihood of missed diagnoses.

The F1 Score balances precision and recall by computing their harmonic mean. It offers a single metric that accounts for both false positives and false negatives, making it useful for evaluating the model's overall performance.

The F1 Score is expressed as:

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

An optimal F1 score indicates a balance between precision and recall, which is essential for an effective diagnostic tool in clinical settings. Achieving this balance ensures the AI model accurately identifies true positives while minimizing false positives and negatives (Appendix A.6). Comprehensive evaluation is critical for clinical adoption, affecting the reliability and trustworthiness of the diagnostic tool.

A.6. Evaluation metrics and accuracy analysis

False positive and negative metrics assessed clinician recognition performance, derived directly from the classification process. These metrics are crucial in medical imaging classification tasks. Their importance is heightened in breast cancer diagnosis, impacting patient care significantly.

False positive and negative metrics provide a more balanced and comprehensive evaluation of performance than classification accuracy, particularly in the presence of imbalanced datasets. For instance, a clinician's BIRADS rating of 3 when the actual BIRADS is 5 constitutes a false-negative result. Conversely, a BIRADS rating of 4 when the actual BIRADS is 2 constitutes a false-positive result, with the “real” score being the ground-truth provided by the HFF expert.

The primary goal was to assess the system's ability to reduce false positives and negatives. Our classifiers achieved an average reduction of approximately 23% in the false positive rate and about 3% in the false negative rate (Section 6.5), outperforming previous approaches (Calisto et al., 2022b). The selected metrics were intended to effectively demonstrate the potential of the AI-assisted approach in minimizing both false positives and false negatives, thereby offering a more dependable diagnostic tool.

The significant reduction in false positives and false negatives highlights the robustness of our approach and indicates the potential for substantial improvements in diagnostic accuracy, when comparing the “real” BIRADS and the provided one (Appendix A.7). These findings enhance confidence in the efficacy of our methods. Furthermore, they pave the way for future research aimed at refining AI models for optimal performance in real-world clinical settings.

A.7. Clustering rationale & diagnostic error

The rationale for using BIRADS scores is foundational to our approach. The BIRADS score ranges from 0 to 6, with the following meanings: 0 – inconclusive, 1 – no findings, 2 – benign findings, 3 – probably benign, 4 – suspicious findings, 5 – high suspicious malignancy, 6 – previously known lesion. BIRADS scores of 0 and 6 are excluded in our study as they do not contribute to prediction purposes.

We clustered the BIRADS scores into three classes to address medical and diagnostic errors:

- “No Findings” with BIRADS = 1
- “Benign, probably benign findings” with BIRADS = {2, 3}
- “Probably, highly suspicious malignancy findings” with BIRADS = {4, 5}

The DenseNet (for 2D MG and US) and the ResNet (for 3D MRI) models were utilized to classify these three classes. We focused on BIRADS scores from 1 to 5 as they are most relevant for diagnosis. Both networks were trained using BIRADS ground truth provided by a radiologist from the HFF clinical institution.

Diagnostic errors were defined based on discrepancies between the provided BIRADS and the actual BIRADS:

- A false negative occurs when the provided BIRADS is lower than the actual BIRADS. For example, if the real BIRADS is 5 (high suspicious malignancy) and the provided BIRADS is 4 (suspicious findings) or 3 (probably benign), it is a false negative.
- A false positive occurs when the provided BIRADS is higher than the actual BIRADS. For example, if the real BIRADS is 1 (no findings) and the provided BIRADS is 2 (benign findings) or 3 (probably benign), it is a false positive.

Our clustering approach enables systematic evaluation and reduction of false positives and negatives. Grouping BIRADS scores into three distinct classes allows for a more effective analysis of diagnostic error patterns and identifies improvement areas for both the AI model and clinicians. This method ensures a thorough evaluation by focusing on critical diagnostic categories, facilitating targeted enhancements in the AI model's performance. Moreover, clustering refines decision-making by providing more precise distinctions between benign and malignant findings. Consequently, this approach enhances our AI-assisted diagnostic tools' diagnostic accuracy and reliability, resulting in more consistent and trustworthy outcomes in clinical practice.

A.8. Enhancing explanations

In this paper, we employ two primary tones for the AI recommendations: (1) assertive, which conveys an authoritative tone, and (2) non-assertive, which adopts a more suggestive tone. Considering the clinical context, future research should not only include more trials but also explore a broader range of communication tones. Explanations should be closely tied to the lexicon concepts for each breast imaging modality (Spak et al., 2017). For example, explanations could include terms like “scattered areas” (Lex_1), “fibroglandular density” (Lex_2), “scattered fibroglandular tissue” (Lex_3), and “ring enhancement mass” (Lex_4). We acknowledge that other communication tones may be beneficial in a clinical context and plan to investigate these in future research.

To comprehensively study the impact of tone styles, we propose the following trials:

1. Conventional Agent;
2. Agent with Explanations in Neutral Tone;
3. Agent with Explanations in Non-Assertive Tone; and
4. Agent with Explanations in Assertive Tone.

This is a promising research direction we are currently pursuing. Providing more detailed and precise explanations will enhance the diagnostic performance of AI-assisted medical imaging classification in the clinical domain of breast cancer.

A.9. Human-AI collaboration in diagnostics

The integration of AI into clinical workflows goes beyond technical accuracy. Effective collaboration between clinicians and AI systems is critical for enhancing diagnostic performance and fostering user trust. Intelligent agents should seamlessly interact with clinicians, allowing them to confidently interpret and validate AI-generated insights.

A.9.1. Supporting expertise and trust

AI systems are required to complement clinical expertise rather than replace it. For less experienced clinicians, AI offers guidance to build confidence in decision-making. AI provides collaborative suggestions for experienced professionals that respect their autonomy and expertise. Trust is essential for effective collaboration. Transparent AI outputs tailored to clinical needs clarify predictions, reduce cognitive load, and support focus on critical decisions.

A.9.2. Collaborative decision-making

AI tools must adapt to diverse clinical environments, integrating smoothly into existing workflows without introducing inefficiencies. Context-sensitive recommendations, flexible communication styles, and modular designs are essential for adoption and utility. Continuous feedback loops further refine AI performance, aligning systems with real-world needs. Collaboration between clinicians and AI should evolve into a symbiotic relationship, leveraging AI's ability to analyze large datasets alongside clinicians' contextual judgment and empathy. Structured workflows that blend these strengths ensure that diagnostic decisions are accurate and informed by clinical expertise and patient-specific nuances.

A.10. Ethical considerations

The adoption of AI in healthcare raises significant ethical considerations, particularly in high-stakes domains like breast cancer diagnostics. Addressing these issues ensures that AI systems contribute to equitable, reliable, patient-centered care.

A.10.1. Mitigating bias in AI models

AI systems are susceptible to biases in training data, which can lead to unequal outcomes across diverse patient populations. For instance, underrepresenting certain demographic groups in imaging datasets may affect the model's reliability for those populations. Developing robust, representative datasets and auditing model performance across subgroups are critical to mitigating bias.

A.10.2. Ensuring accountability

The introduction of AI complicates accountability in diagnostic workflows. While AI systems provide recommendations, clinicians are ultimately responsible for patient outcomes. Designing systems that clarify the division of responsibility while empowering clinicians with tools to override or validate AI outputs ensures accountability remains intact.

A.10.3. Protecting patient privacy

AI systems process large volumes of sensitive medical data, requiring strict privacy safeguards. Compliance with regulations like GDPR or HIPAA and advanced data encryption and anonymization techniques is essential to protect patient confidentiality while enabling robust AI training and evaluation.

A.10.4. Addressing ethical trade-offs

Balancing precision and recall in AI models often involves ethical trade-offs. High Recall reduces missed diagnoses, but higher false positives can lead to unnecessary interventions and patient anxiety. Engaging clinicians, ethicists, and patients in decision-making ensures these trade-offs align with shared values and clinical priorities.

A.10.5. Promoting equitable access

AI systems must be accessible across diverse clinical settings, from resource-rich hospitals to underfunded healthcare facilities. Reducing cost barriers and simplifying deployment processes ensures that AI benefits are distributed equitably, contributing to global improvements in breast cancer care.

Appendix B. Repositories

Our repositories are accessible to the public and can be easily located online. Please follow the [prototype-assertive-reactive](https://github.com/MIMBCD-UI/prototype-assertive-reactive) repository (github.com/MIMBCD-UI/prototype-assertive-reactive) for more details about the source code of the prototypes. In terms of results and statistical analysis, all information is available in the [sa-uta11-results](https://github.com/MIMBCD-UI/sa-uta11-results) repository (github.com/MIMBCD-UI/sa-uta11-results). Links were accessed on July 23, 2024.

Appendix C. Intellectual property

The work described in this paper is covered by grating rights of two patent applications (Calisto and C. Nascimento, 2020; Calisto et al., 2023b), filed by Instituto Superior Técnico. The first patent is titled “Computational Method and System for Improved Identification of Breast Lesions” and is already internationally granted in several regions (Calisto and C. Nascimento, 2020). The second patent is titled “Machine Learning System for Diagnosing Breast Lesions, Adaptable to the Clinician and Based on Assertiveness” and is under pending status (Calisto et al., 2023b). The contents of this paper aim to inform the scientific and technical community without limiting the scope of the pending patent application. Patent rights will be enforced as needed to protect the patent holders' proprietary interests. For more information, further details are available in the [LICENSE.md](https://github.com/MIMBCD-UI/sa-uta11-results) file of the [sa-uta11-results](https://github.com/MIMBCD-UI/sa-uta11-results) repository (github.com/MIMBCD-UI/sa-uta11-results). Accessed on July 23, 2024.

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Data availability

Data and code for this research are publicly available on a GitHub repository. The link to the repository has been shared. Please follow the link: <https://github.com/MIMBCD-UI/sa-uta11-results>.

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